STRATIFIED FUNDAMENTAL-GROUP COHOMOLOGY OF PROJECTIVE CURVES AGREES WITH STRATIFIED COHOMOLOGY AND VANISHES ABOVE DEGREE ONE

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Allowed standard prerequisites (editorial).

- Tannakian groups, stratified/Frobenius-divided sheaves, rational group cohomology and limit functors. Original complete author formulations, limit arguments and bibliography retained.

1. INTRODUCTION

Let X be a smooth geometrically connected algebraic variety over an algebraically closed field k of characteristic $p > 0$. A *stratified sheaf* is a quasi-coherent $\mathscr{O}_X$-module equipped with an action of the sheaf of differential operators $\mathscr{D}(X/k)$. The terminology "stratified" is due to Grothendieck, in connection with the infinitesimal cohomology, see, e.g., [BO78, Section 2] for more details. It is shown that a stratified sheaf which is coherent as an $\mathscr{O}_X$-module is automatically locally free, hence will be referred to as a "stratified bundle". As a consequence, the category of stratified bundles is an abelian, rigid tensor category.

The category of stratified sheaves is denoted by $\mathrm{str}(X/k)$, the full subcategory of stratified bundles is denoted by $\mathrm{str}^{\mathrm{coh}}(X/k)$. The cohomology of stratified sheaves as the right derived functors of the hom-set functor in $\mathrm{str}(X/k)$ was studied by Ogus in [Ogu75], see also [Hai13].

An action of $\mathscr{D}(X/k)$ on a sheaf $\mathscr{E}$ determines a descending chain of subsheaves $\mathscr{E}^{(n)}$ which are obtained from one another by the Frobenius pull-back: $\mathscr{E}^{(n)} \cong F^*\mathscr{E}^{(n+1)}$. Such a sequence $(\mathscr{E}^{(n)})$ together with the isomorphisms $\sigma_n : \mathscr{E}^{(n)} \longrightarrow F^*\mathscr{E}^{(n+1)}$ gives an alternative definition for a stratified sheaf, in case $\mathscr{E}$ is coherent, $(\mathscr{E}^{(n)}, \sigma_n)$ is called a *flat bundle*, cf. [Gie75, Section 1]. Some other authors also call it an *F-divided bundle* to generalize it to a more general setting, see, e.g., [BHdS23].

Let x be a k-point of X. The fiber functor at x:

$$x^* : \mathrm{str}^{\mathrm{coh}}(X/k) \longrightarrow \mathrm{Vec}_k, \quad (\mathscr{V}, \nabla) \mapsto \mathscr{V}_x,$$

is faithfully exact (since the kernel and cokernel of a map in $\mathrm{str}^{\mathrm{coh}}(X)$ are always locally free). Hence, Tannakian duality (cf. [DM82, Theorem 2.11]) yields an affine group scheme, denoted by $\pi(X, x)$, with the property that x^* gives an equivalence:

$$x^* : \mathrm{str}^{\mathrm{coh}}(X/k) \xrightarrow{\cong} \mathrm{Rep}^{\mathrm{f}}(\pi(X, x)). \tag{1}$$

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This equivalence extends to an equivalence between the ind-category $\mathrm{str}^{\mathrm{ind}}(X/k)$ and the category $\mathrm{Rep}(\pi(X,x))$. Here $\mathrm{str}^{\mathrm{ind}}(X/k)$ denotes the category of stratified sheaves which can be presented as inductive limits of stratified bundles. We shall refer to them as *ind-stratified bundles*. The group $\pi(X,x)$ can be seen as an analog in positive characteristic of the differential fundamental group. It was implicitly studied in the work [Gie75] of Gieseker and more recently in the work of dos Santos [dSa07] and subsequently by many authors, see, e.g., [EM10, Kin15, ESB16, vdP19, Sun19].

In what follows we shall fix the point x and abbreviate $\pi(X,x)$ to $\pi(X)$ for simplicity.

A central theme of this work is the interaction between the cohomology theory of stratified sheaves and the rational cohomology attached to the stratified fundamental group. Although Tannakian duality provides an equivalence between stratified bundles and finite dimensional representations of the stratified fundamental group, the corresponding cohomology theories are a priori defined in rather different ambient categories. In particular, the stratified cohomology is computed inside the larger category of all stratified sheaves and is governed by inverse systems arising from Frobenius descent, while the group cohomology is defined through injective resolutions in the representation category of an affine group scheme. Understanding the precise relationship between these two theories therefore requires a careful analysis of how Frobenius-divided structures behave under limits and extensions. More precisely, the equivalence in (1) implies that the group cohomology of $\pi(X)$ with coefficients in a representation V is the same as the cohomology of the corresponding stratified bundle $\mathcal{V}$ computed in the category $\mathrm{str}^{\mathrm{ind}}(X/k)$ (that is the derived functors of the hom-set functor computed in $\mathrm{str}^{\mathrm{ind}}(X/k)$). Consequently, there is a natural map

$$\delta^i : \mathrm{H}^i(\pi(X), V) \longrightarrow \mathrm{H}^i_{\mathrm{str}}(X, \mathcal{V}), \tag{2}$$

where $\mathcal{V}$ is a stratified bundle and V its fiber at x, since the functors on both sides of this natural map are derived functors of the hom-set functors in the corresponding categories. In particular δ^0 is an isomorphism. Further, as extensions of vector bundles are again vector bundles, δ^1 is also an isomorphism.

However, $\mathrm{str}^{\mathrm{ind}}(X/k)$ is much smaller than the ambient category $\mathrm{str}(X/k)$. Therefore, the maps δ^i are not expected to be injective or surjective. Our first main finding is that, for curves, δ^i, $i \geq 0$, are nevertheless bijective.

Theorem 1 (Theorem 3.1.1)**.** *For a smooth projective curve X over k the comparison maps δ^i in* (2) *are isomorphisms for all $i \geq 0$ and all stratified bundles $\mathcal{V}$. As a consequence, the cohomology of $\pi(X)$ vanishes from degree $i \geq 2$.*

For smooth, complete varieties over an algebraically closed field, the étale fundamental group is the profinite quotient of the stratified fundamental group. As a consequence, we recover the following results :

(1) (Proposition 3.2.1) The restriction map $\rho^i : \mathrm{H}^i(\pi^{\text{ét}}(X), V) \longrightarrow \mathrm{H}^i(\pi(X), V)$ given by the canonical surjective map $\pi(X) \longrightarrow \pi^{\text{ét}}(X)$, is bijective for any $i \geq 0$ and any representation V of $\pi^{\text{ét}}(X)$. As a consequence, the cohomology of $\pi^{\text{ét}}(X)$ vanishes from degree $i \geq 2$.

(2) (Corollary 3.3.1, Proposition 3.3.6) Let $\pi^{\mathrm{uni}}(X)$ be the Tannakian dual to the full subcategory of nilpotent stratified bundles. Then $\mathrm{H}^i(\pi^{\mathrm{uni}}(X), V) \cong \mathrm{H}^i(\pi(X), V)$ As a consequence, the cohomology of $\pi^{\mathrm{uni}}(X)$ vanishes from degree $i \geq 2$ so that $\pi^{\mathrm{uni}}(X)$ is a free pro-étale, pro-unipotent group scheme.

The property (1) is due to Serre [Ser90] and Pacheco-Stevenson [PS00].

From a conceptual point of view, our comparison theorem suggests that smooth projective curves in positive characteristic should be regarded as "stratified $K(\pi,1)$-schemes" (cf. [BHT25, HBB26, BN26] and references therein). Namely, the entire stratified cohomology of coherent stratified bundles is recovered from the group cohomology of the stratified fundamental group. This phenomenon is reminiscent of the classical topological situation where the cohomology of local systems on a $K(\pi,1)$-space is completely determined by the cohomology of its fundamental group.

We note an important difference: the stratified cohomology groups vanish already in degrees at least 2, that is, stratified cohomology does not satisfy the property of a Weil cohomology. This differs from the situation for the differential fundamental group in characteristic 0 (cf. [BHT25]) where the cohomology completely vanishes only from degree larger than 2. Such a phenomenon was already known in the work of Gieseker, who showed the "rigidity" of regular singular stratified bundles in the formal neighborhood of a point [Gie75, Theorem 3.3].

Motivated by the non-vanishing of the first de Rham cohomology of smooth projective curves of genus $g > 0$ [BHT25], we ask a similar question for stratified cohomology of curves in positive characteristic. It turns out that, unlike the de Rham cohomology for curves in characteristic 0, in positive characteristic, the first stratified cohomology group may vanish. To see that, we carry out some computations for finite stratified bundles, that is étale trivializable vector bundles. Our explicit computation on the curves in Section 4.2 also shows that the non-vanishing of stratified cohomology in degree 1 depends on some invariants of the curves. This point of view suggests many interesting questions, which we hope to address in the future. Our second main finding is the following.

Proposition 2 (Corollary 3.5.4 and Appendix A.6)**.** (1) *Let X be an ordinary curve of genus larger than* 1. *Let $\mathscr{E}$ be an ordinarily étale trivializable vector bundle. Then* $\mathrm{h}^1_{\mathrm{str}}(X, \mathscr{E}) \neq 0$.

(2) *There exists an étale trivializable vector bundle on a hyperelliptic curve for which the first stratified cohomology is equal to zero.*

Another important aspect of the paper is that the comparison theorem may fail for ind-stratified bundles. Already on an ordinary elliptic curve, an infinite sum of rank-one stratified bundles can produce a non-zero derived inverse limit and hence a failure of comparison; see Example 4.1.2. For curves of higher genus we exhibit a different phenomenon. We construct explicit Frobenius invariant iterated extensions on hyperelliptic curves (see Section 4) and introduce the notion of Cartier nilpotency order (cf. Proposition 4.3.3). The nilpotency order can grow unboundedly along a sequence of extensions, leading to an ind-stratified bundle for which the natural comparison map is no longer bijective even though the whole construction stays in the unipotent isotypical component. These examples highlight the essential difference between coherent and ind-coherent phenomena. They also emphasize that, while group cohomology commutes with inductive limits, stratified cohomology is computed in terms of inverse limit functors (see Section 2.2), which need not commute with direct limits.

Our main finding in this direction is the following.

Proposition 3 (Proposition 4.3.4 and Lemma 4.4.6)**.** *Let X be a hyperelliptic curve of genus 2 and Hasse-Witt rank 1 over an algebraically closed field k of characteristic* 3*. There exists an ind-stratified bundle $\mathscr{E}$, which is an infinite extension of the trivial stratified bundle, such that the canonical map*

$$\delta^1 : \mathrm{H}^1(\pi(X), E) \longrightarrow \mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E})$$

is not bijective. In fact, the target space is infinite dimensional while the source space is zero.

Let us briefly describe the structure of our work. In Section 2, we recall the definitions of stratified bundles and the stratified cohomology following Gieseker [Gie75] and Ogus [Ogu75]. Each stratified bundle $\mathscr{E}$ can be equivalently given as a sequence of vector bundles $\mathscr{E}^{(n)}$ such that $F^*\mathscr{E}^{(n+1)} \cong \mathscr{E}^{(n)}$, F being the absolute Frobenius map. The i-th stratified cohomology is computed from the inverse limit of the pro-system $\mathrm{H}^i(X, \mathscr{E}^{(n)})$, in which the transfer maps are given by the action of the Frobenius. Thus, the computation of stratified cohomology is reduced to the study of the Frobenius action on the cohomology groups of sequence of vector bundles related to each other by the Frobenius pull-back. We show the finiteness and vanishing properties of this cohomology, cf. Proposition 2.2.2.

In Section 3, we define a natural map from the group cohomology of the stratified fundamental group to the stratified cohomology. We show that for smooth projective curves the maps are bijective, cf. Theorem 3.1.1. Since stratified cohomology on projective curves vanishes in degree larger than 1, we obtain a similar result for the fundamental group cohomology. This vanishing theorem also allows us to compare the cohomology of the stratified fundamental group with the cohomology of the étale fundamental group and its pro-unipotent quotient in k-linear representations (Proposition 3.2.1 and Corollary 3.3.1). A key ingredient here is the fact, due to dos Santos [dSa07], that any unipotent quotient of the stratified fundamental group is finite. In Subsection 3.2 we compare the stratified cohomology of finite stratified bundles with the étale cohomology of their p-torsion sub-sheaves of Frobenius invariant sections. Then we compare the stratified cohomology of finite stratified bundles with cohomology of the corresponding p-torsion étale sheaves. Our motivation is to study the non-vanishing of the first cohomology group.

In Section 4, we consider the case of ind-stratified bundles. For a supersingular elliptic curve the comparison maps δ^i remain bijective for arbitrary ind-stratified bundles. For an ordinary elliptic curve the same holds for isotypical ind-stratified bundles and their finite direct sums, but it can fail for infinite sums of isotypical components; we give a short explicit counterexample. Section 4.2 introduces the notion of Cartier nilpotency of Frobenius invariant vector bundles. We construct, on a hyperelliptic curve of genus $g = 2$, a sequence $\mathscr{E}_n$ of iterated extensions of the trivial bundles which are Frobenius invariant. Under the abstract Cartier condition (C) at $\mathscr{E}_3$, we show that the order of Cartier nilpotency of $\mathscr{E}_n$ tends to infinity with n. Section 4.4 verifies this condition on explicit curves in characteristic three. As a consequence, the first stratified cohomology of $\mathscr{E} = \varinjlim \mathscr{E}_n$ is infinite dimensional and hence the comparison map δ^1 is not bijective. In contrast with the ordinary elliptic example, this second failure occurs entirely inside the unipotent isotypical component.

2. Stratified bundles and stratified cohomology

2.1. **Stratified bundles.** Let k be an algebraically closed field of characteristic $p>0$ and let X/k be a smooth, geometrically connected scheme.

2.1.1. Let $\mathscr{D}(X/k)$ be the sheaf of differential operators on X/k, cf. [EGA4, IV.16.8]. A stratified sheaf on X/k is a quasi-coherent $\mathscr{O}_X$-module $\mathscr{V}$ equipped with an action of $\mathscr{D}(X/k)$, that is, a k-linear map of sheaves of algebras

$$\nabla : \mathscr{D}(X/k) \longrightarrow \mathscr{E}nd_k(\mathscr{V}).$$

∇ is also called a stratification on $\mathscr{V}$. The category of stratified sheaves over X/k is denoted by $\mathrm{str}(X/k)$. We denote by $\mathrm{str}^{\mathrm{coh}}(X/k)$ the full subcategory of $\mathscr{O}_X$-coherent stratified sheaves. According to a theorem of Grothendieck, an $\mathscr{O}_X$-coherent module equipped with a stratification is always locally free [BO78, Prop. 2.16] or [dSa07, Lemma 6]. We shall therefore address objects of $\mathrm{str}^{\mathrm{coh}}(X/k)$ as stratified bundles.

Thus $\mathrm{str}^{\mathrm{coh}}(X/k)$ is a rigid tensor category with the unit object being the trivial connections $(\mathscr{O}_X, d)$ where d denotes the natural inclusion of $\mathscr{D}(X/k)$ into $\mathscr{E}nd_k(\mathscr{O}_X)$.

2.1.2. Let $(\mathscr{E},\nabla)$ be a stratified sheaf on X/k. We define $\mathscr{E}^{(i)}$ the subsheaf of sections of $\mathscr{E}$ annihilated by differential operators of order less than p^i and vanishing on constant functions. In other words, let $\mathscr{D}^{<p^i}$ be the subsheaf of differential operators of order less than p^i. Then

$$\mathscr{E}^{(i)} = \mathscr{H}om_{\mathscr{D}^{<p^i}}(\mathscr{O}_X, \mathscr{E}),$$

in particular, $\mathscr{E}^{(0)} = \mathscr{E}$. We define the sheaf of horizontal sections of $(\mathscr{E},\nabla)$ to be

$$\mathscr{E}^{\nabla} := \bigcap_i \mathscr{E}^{(i)} = \varprojlim_i \mathscr{E}^{(i)}.$$

We have

$$\mathrm{Hom}_{\mathrm{str}}((\mathscr{O}_X, d), (\mathscr{E},\nabla)) = \mathrm{H}^0(X, \mathscr{E}^{\nabla}) = \bigcap_i \mathrm{H}^0(X, \mathscr{E}^{(i)}) = \varprojlim_i \mathrm{H}^0(X, \mathscr{E}^{(i)}),$$

here $\mathrm{Hom}_{\mathrm{str}}$ is the abbreviation of $\mathrm{Hom}_{\mathrm{str}(X/k)}$. This space is always finite dimensional, cf. [Hai13, Lemma 1.5] or 3.1 below.

2.1.3. According to a result of Cartier, the Frobenius pull-back of $\mathscr{E}^{(1)}$ is isomorphic to $\mathscr{E}$, that is there is an isomorphism

$$\sigma_0 : F^*\mathscr{E}^{(1)} \cong \mathscr{E},$$

where $F : X \longrightarrow X$ is the absolute Frobenius map, which on the structure sheaf is given by $F : \mathscr{O}_X \longrightarrow \mathscr{O}_X$, $s \mapsto s^p$. There is a p-linear map from $\mathscr{E}^{(1)}$ to $F^*\mathscr{E}^{(1)}$. Composing it with σ_0 we obtain a p-linear map $\psi_1 : \mathscr{E}^{(1)} \longrightarrow \mathscr{E}$.

More generally, the Frobenius pull-back of $\mathscr{E}^{(i+1)}$ is isomorphic to $\mathscr{E}^{(i)}$. As a consequence, there is an alternative definition of stratified sheaves as infinite sequences of sheaves $(\mathscr{E}^{(i)})_{i\geq 0}$ together with isomorphisms $\sigma_i : F^*\mathscr{E}^{(i+1)} \cong \mathscr{E}^{(i)}$, for all $i \geq 0$, cf. [Gie75]. This data is called a flat bundle by Gieseker. We shall denote by ψ_n the resulting p^n-linear map $\mathscr{E}^{(n)} \longrightarrow \mathscr{E}$. We notice that, on a projective smooth variety, the stratified bundle associated to a flat bundle is (up to an isomorphism) independent of the choice of the isomorphism σ_i [Gie75, Lemma 1.7].

2.1.4. We notice that $\mathrm{str}^{\mathrm{ind}}(X/k)$ is strictly smaller than $\mathrm{str}(X/k)$. For instance, consider $\mathscr{D}(X/k)$ as a stratified sheaf on itself, then it cannot be presented as the union of its coherent stratified subsheaves, since any proper subsheaf cannot contain the unit section.

2.2. **Stratified cohomology.** When it is clear which stratification is referred to, we shall omit the symbol for it in the notation of a stratified sheaf, thus we shall write simply $\mathscr{E}$ instead of $(\mathscr{E},\nabla)$. The 0-th stratified cohomology group of a stratified sheaf $\mathscr{E}$ is defined to be

$$\mathrm{H}^0_{\mathrm{str}}(X/k,\mathscr{E}) := \mathrm{Hom}_{\mathrm{str}}(\mathscr{O},\mathscr{E}) = \mathrm{H}^0(X,\mathscr{E}^{\nabla}),$$

where $\mathscr{O}$ is equipped with the trivial stratification 2.1.1. This is a left exact functor from $\mathrm{str}(X/k)$ to the category of k-vector spaces. We notice that $\mathrm{str}(X/k)$ has enough injectives. Hence, we can define the derived functors $\mathrm{H}^i_{\mathrm{str}}(X/k,-)$, which are the higher stratified cohomology functors. In particular, we have

$$\mathrm{H}^i_{\mathrm{str}}(X/k,-) \cong \mathrm{Ext}^i_{\mathrm{str}}(\mathscr{O}_X,-).$$

2.2.1. *Computation.* Following Ogus (cf. [Ogu75]), we provide some computations of the stratified cohomology in terms of the inverse limits. As in 2.1.2, the functor $\mathrm{H}^0_{\mathrm{str}}(X/k,-)$ can be seen as the composed functor

$$\mathrm{H}^0_{\mathrm{str}}(X/k,\mathscr{E}) = \varprojlim_n \mathrm{H}^0(X,\mathscr{E}^{(n)}).$$

Therefore, there is a spectral sequence abutting to it. As a consequence, we have exact sequences (cf. [Ogu75, Theorem 2.4]) for any $i \geq 1$:

$$0 \to \varprojlim_n{}^1 \mathrm{H}^{i-1}(X,\mathscr{E}^{(n)}) \to \mathrm{H}^i_{\mathrm{str}}(X/k,(\mathscr{E},\nabla)) \to \varprojlim_n \mathrm{H}^i(X,\mathscr{E}^{(n)}) \to 0.$$

2.2.2. Recall that the limit on inverse systems of vector spaces (or modules) is a left exact functor. Its derived functors, except possibly for the first one, vanish. Further, for an inverse system $(X_i, t_i : X_{i+1} \longrightarrow X_i)_{i\geq 0}$ of vector spaces (or modules over a ring), we have an exact sequence

$$0 \longrightarrow \varprojlim X_i \longrightarrow \prod_i X_i \xrightarrow{d} \prod_i X_i \longrightarrow \varprojlim{}^1 X_i \longrightarrow 0,$$

where the map d sends a tuple (x_i) to the tuple $(x_i - t_i(x_{i+1}))$. As a consequence, $\varprojlim{}^1(X_i) = 0$ if and only if the map d is surjective. And we have the following sufficient condition for that.

Proposition 2.2.1. *If X_i is complete and separated with respect to the filtration*

$$F^k(X_i) = \mathrm{Im}(X_{i+k} \longrightarrow X_i)$$

for every i, then d is surjective.

Proof. For $j \geq i$, write

$$t_{i,j} := t_i \circ t_{i+1} \circ \cdots \circ t_j : X_{j+1} \longrightarrow X_i.$$

Given $y = (y_i)_i \in \prod_i X_i$, consider the partial sums

$$x_i^{(N)} := y_i + \sum_{j=i}^{N} t_{i,j}(y_{j+1}) \qquad (N \geq i).$$

Their tails lie in arbitrarily deep terms of the filtration on X_i. Completeness and separatedness therefore give a well-defined limit $x_i = \lim_N x_i^{(N)} \in X_i$. Passing to the limit in $x_i^{(N)} - t_i(x_{i+1}^{(N)}) = y_i$ (for $N \geq i+1$) gives $d(x) = y$.

The preimage x need not be unique: any two preimages differ by an element of $\ker(d) = \varprojlim_i X_i$. Separatedness is used for uniqueness of the limit of the partial sums, rather than for uniqueness of a preimage under d. □

In particular, if the spaces X_i are finite dimensional, then for each i the images of $X_j \to X_i$ stabilize as $j \to \infty$. Thus the system satisfies the Mittag–Leffler condition and its first derived limit vanishes. Moreover, if the dimensions of the spaces X_i are uniformly bounded (i.e., $\dim(X_i)$ is bounded above by a constant c for all i), then $\varprojlim X_i$ is finite-dimensional.

Proposition 2.2.2. *Let X be a smooth connected projective variety over k, and $\mathscr{E} = \{\mathscr{E}^{(n)}\}$ be a stratified bundle. Then the stratified cohomology group can be computed as the inverse limit of the Zariski cohomology of $\mathscr{E}^{(n)}$:*

$$\mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{E}) = \varprojlim_n \mathrm{H}^i(X, \mathscr{E}^{(n)}).$$

Thus $\mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{E})$ are finite dimensional and vanish for $i > \dim X$.

Proof. We show that the cohomology groups $\mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{E})$ are finite dimensional for all i and $\mathscr{E}$. For this, we first recall some properties of the Hilbert polynomial of $\mathscr{E}$. Let $\mathscr{O}_X(1)$ be an ample line bundle. For a bundle B of rank $r > 0$, set $B(m) = B \otimes_{\mathscr{O}_X} \mathscr{O}_X(m)$, and define the Hilbert polynomial of B relative to $\mathscr{O}_X(1)$ to be

$$p_B(m) = \frac{1}{r}\chi(X, B(m)) \in \mathbb{Q}[m] \quad \text{for } m \gg 0,$$

Additionally, we define the slope of B to be $\mu(B) = \deg(B)/r$. B is said to be semistable (or μ-semistable) if for all subsheaves $U \subset B$ we have $\mu(U) \leq \mu(B)$.

Consider the stratified bundle $\mathscr{E} = (\mathscr{E}^{(n)}, \sigma_n)_{n\in\mathbb{N}}, \sigma_n : F^*\mathscr{E}^{(n+1)} \cong \mathscr{E}^{(n)}$, of rank $r > 0$. Since

$$\mu((F^n)^*\mathscr{G}) = p^n \mu(\mathscr{G}),$$

we see that $\mathscr{E}^{(n)}$ is μ-semistable when n is large enough. Moreover, by [EM10, Corollary 2.2], the Hilbert polynomials of $\mathscr{E}^{(n)}$ are the same and equal to that of $\mathscr{O}_X$. Now Langer's boundedness theorem [La04, Theorem 4.2] tells us that the family $\{\mathscr{E}^{(n)}\}$ is bounded, i.e., there exists a scheme S of finite type over k and an S-flat family $\mathscr{F}$ of coherent sheaves on the fibers of $X \times S \to S$, which contains $\{\mathscr{E}^{(n)}\}$. From the above argument, we have that the family of bundles $\{\mathscr{E}^{(n)}\}_{n\in\mathbb{N}}$ comes from a S-flat coherent sheaf, say $\mathscr{F}$, over $X \times S$, where S is a scheme of finite type over k. Hence, to show the uniformly boundedness of $\mathrm{H}^i(X, \mathscr{E}^{(n)})$, it is enough to consider the case S is affine. In that case, set $S = \mathrm{Spec}(A)$, for some finitely generated k-algebra A. Since X is proper, the projection map $X \times S \to S$ is proper too. Hence, by (the Grothendieck complex), there is a finite complex $K^\bullet$ of finitely generated A-modules such that

$$\mathrm{H}^i(K^\bullet \otimes_A B) = \mathrm{H}^i((X \times \mathrm{Spec}(B), \mathscr{F} \otimes_A B),$$

for any A-algebra B. So we deduce our first claim by taking B to be $k(s)$ for any point $s \in S$. □

Remark 2.2.3. i) We note the fundamental difference between stratified cohomology in positive characteristic and its characteristic-zero counterpart, namely de Rham cohomology. While the de Rham cohomology vanishes only in degrees larger than two times the dimension of the variety, stratified cohomology vanishes already in degrees larger than the dimension of the variety.

ii) In a recent preprint [Xia25], the author also proves the finiteness of the stratified cohomology on a smooth and proper scheme by using a similar method. Since he does not assume the projectivity of X, the proof there is more technical.

3. Comparison with fundamental group cohomology

3.1. **The stratified fundamental group and cohomology-comparison theorem.** From the assumption that X is geometrically connected, we have

$$\mathrm{End}_{\mathrm{str}}((\mathscr{O}_X,d)) = \mathrm{H}^0_{\mathrm{str}}(X,(\mathscr{O}_X,d)) = \bigcap_n (\mathscr{O}_X)^{p^n} = k.$$

On the other hand, as the underlying sheaves of objects of $\mathrm{str}^{\mathrm{coh}}(X/k)$ are locally free, the fiber functor at any point is exact, hence is faithful. Consequently, $\mathrm{str}^{\mathrm{coh}}(X/k)$ is a Tannakian category with respect to the fiber functor at any point $x \in X(k)$. In particular, for any stratified bundle $\mathscr{E}$, the space $\mathrm{H}^0_{\mathrm{str}}(X,\mathscr{E})$ is finite dimensional. Tannakian duality yields an affine group scheme $\pi(X)$ (we shall not mention the base point x in what follows) with the property: the functor x^* yields an equivalence:

$$x^* : \mathrm{str}^{\mathrm{coh}}(X/k) \xrightarrow{\cong} \mathrm{Rep}^{\mathrm{f}}(\pi(X)), \tag{3}$$

which extends to an equivalence

$$x^* : \mathrm{str}^{\mathrm{ind}}(X/k) \xrightarrow{\cong} \mathrm{Rep}(\pi(X)). \tag{4}$$

According to the Tannakian duality, the functor x^* yields an equivalence between $\mathrm{str}^{\mathrm{ind}}(X/k)$ and the representation category of an affine group scheme which is called the stratified fundamental group of X at the base point x and is denoted by $\pi(X,x)$:

$$x^* : \mathrm{str}^{\mathrm{ind}}(X/k) \cong \mathrm{Rep}(\pi(X,x)).$$

We will simplify the notation to $\pi(X)$. The category $\mathrm{Rep}(\pi(X))$ has enough injectives. Hence we can define the cohomology of $\pi(X)$ with coefficients in its representation V to be the right derived functors of the invariant-subspace functor

$$\mathrm{H}^0(\pi(X),V) := \mathrm{Hom}_{\pi(X)}(k,V) = V^{\pi(X)},$$

where k stands for the trivial representation. The i-th cohomology group is denoted, as usual, by $\mathrm{H}^i(\pi(X),V)$. We also have

$$\mathrm{H}^i(\pi(X),V) \cong \mathrm{Ext}^i_{\pi(X)}(k,V).$$

These isomorphisms yield natural maps

$$\delta^i : \mathrm{H}^i(\pi(X),E) \longrightarrow \mathrm{H}^i_{\mathrm{str}}(X/k,\mathscr{E}),$$

where $E := x^*\mathscr{E} = \mathscr{E}|_x$. Since an extension in $\mathrm{str}(X/k)$ of two stratified bundles is again a stratified bundle, the maps δ^0 and δ^1 are isomorphisms for coherent coefficients. We shall use this degree-one comparison only in this finite-dimensional setting below.

Theorem 3.1.1. *Let X be a smooth projective curve over an algebraically closed field k of positive characteristic, $x \in X(k)$. Then for any stratified bundle $(\mathscr{E}, \nabla)$ and $E := \mathscr{E}|_x$, all the maps*

$$\delta^i : \mathrm{H}^i(\pi(X), E) \longrightarrow \mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{E})$$

are bijective, moreover, δ^i for $i \geq 2$ are all zero-maps, that is

$$\mathrm{H}^i(\pi(X), E) = \mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{E}) = 0$$

for all $i \geq 2$.

Proof. The comparison maps in degrees zero and one are isomorphisms, as explained above. Proposition 2.2.2 gives

$$\mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{E}) = 0 \qquad (i \geq 2).$$

It therefore suffices to prove the corresponding vanishing for $\pi(X)$.

We shall use the fact that rational group cohomology commutes with filtered direct limits (see the beginning of Section 4).

Choose an exact sequence

$$0 \longrightarrow E \longrightarrow J \longrightarrow Q \longrightarrow 0$$

with J injective in $\mathrm{Rep}(\pi(X))$. Write Q as the filtered union of its finite-dimensional subrepresentations Q_α, and let $J_\alpha \subset J$ be the inverse image of Q_α. Then

$$0 \longrightarrow E \longrightarrow J_\alpha \longrightarrow Q_\alpha \longrightarrow 0$$

is exact, and $\dim_k J_\alpha = \dim_k E + \dim_k Q_\alpha < \infty$. Under Tannakian duality, it corresponds to an exact sequence of coherent stratified bundles

$$0 \longrightarrow \mathscr{E} \longrightarrow \mathscr{J}_\alpha \longrightarrow \mathscr{Q}_\alpha \longrightarrow 0.$$

Since $\mathrm{H}^2_{\mathrm{str}}(X/k, \mathscr{E}) = 0$, the associated long exact sequence and the natural degree-one comparison give a surjection

$$\mathrm{H}^1(\pi(X), J_\alpha) \twoheadrightarrow \mathrm{H}^1(\pi(X), Q_\alpha).$$

Here the comparison is applied only to the coherent bundles $\mathscr{J}_\alpha$ and $\mathscr{Q}_\alpha$.

We have $J = \varinjlim_\alpha J_\alpha$ and $Q = \varinjlim_\alpha Q_\alpha$. Taking filtered direct limits of these surjections therefore yields

$$0 = \mathrm{H}^1(\pi(X), J) \twoheadrightarrow \mathrm{H}^1(\pi(X), Q),$$

where the vanishing on the left follows from injectivity of J. The long exact sequence of $0 \to E \to J \to Q \to 0$ now gives

$$\mathrm{H}^2(\pi(X), E) \cong \mathrm{H}^1(\pi(X), Q) = 0.$$

This proves degree-two vanishing for every finite-dimensional representation. Every rational $\pi(X)$-representation is a filtered union of finite-dimensional subrepresentations, so the same commutation with filtered direct limits gives $\mathrm{H}^2(\pi(X), V) = 0$ for every $V \in \mathrm{Rep}(\pi(X))$. For such a V, choose $0 \to V \to I \to W \to 0$ with I injective. Dimension shifting gives

$$\mathrm{H}^i(\pi(X), V) \cong \mathrm{H}^{i-1}(\pi(X), W) \qquad (i \geq 2).$$

Induction on i, starting with degree two and applying the assertion to every rational representation, proves $\mathrm{H}^i(\pi(X), V) = 0$ for all $i \geq 2$. Taking $V = E$ completes the proof. □

3.2. **Comparison with cohomology of the étale fundamental group.** We call a vector bundle on X *étale finite* if it can be trivialized by an étale finite torsor (i.e. torsor under an étale finite group scheme) on X. The category $\mathscr{C}^{\text{ét}}(X)$ of étale finite vector bundles equipped with the fiber functor x^* is Tannakian (the fixed $k-$ point x will be omitted) and its dual is precisely the (geometric) étale fundamental group scheme $\pi^{\text{ét}}(X)$ (considered as a pro-finite group scheme). The reader is referred to [No82], [MS02] and [EHS08] for more details. Further, there exists a natural functor

$$\mathscr{C}^{\text{ét}}(X) \longrightarrow \text{str}(X/k)$$

in the sense that each étale finite vector bundle is equipped in a canonical way a stratification making it a stratified bundle with finite monodromy (cf. [dSa07]). This functor is fully faithful, exact and closed under taking subquotients (which is equivalent to the surjectivity of the map $\pi(X) \longrightarrow \pi^{\text{ét}}(X)$). For convenience, we shall identify $\mathscr{C}^{\text{ét}}(X)$ with the full subcategory of stratified bundle with finite monodromy $\text{str}^{\text{fin}}(X/k)$.

We notice that the condition "étale finite" is equivalent to the condition "étale trivializable" as defined in [LS77]. We shall use interchangeably these notations.

Let $\text{str}^{\text{uni}}(X)$ be the full subcategory of $\text{str}^{\text{coh}}(X)$ of objects which are iterated extension of the trivial object $(\mathscr{O}, d)$. The corresponding Tannakian group is a pro-unipotent group, denoted $\pi^{\text{uni}}(X)$. It is a quotient group scheme of $\pi(X)$, and, according to [dSa07], it is pro-finite, hence is a quotient of $\pi^{\text{ét}}(X)$.

Summarizing, we have surjective (i.e. faithfully flat) maps

$$\pi(X) \longrightarrow \pi^{\text{ét}}(X) \longrightarrow \pi^{\text{uni}}(X).$$

Proposition 3.2.1. *Let X/k be a smooth projective curve. For any finite representation V of $\pi^{\text{ét}}(X)$ the natural maps induced by restriction*

$$\text{res}^i : \text{H}^i(\pi^{\text{ét}}(X), V) \longrightarrow \text{H}^i(\pi(X), V)$$

are isomorphisms.

Proof. The claim for $i = 0$ follows from the surjectivity of $\pi(X) \longrightarrow \pi^{\text{ét}}(X)$. The surjectivity also ensures the injectivity of res^1.

The claim for $i = 1$ amounts to the thickness of $\mathscr{C}^{\text{ét}}(X)$ in $\text{str}(X/k)$, that is, for any extension

$$0 \longrightarrow \mathscr{V} \longrightarrow \mathscr{W} \longrightarrow \mathscr{O}_X \longrightarrow 0$$

in $\text{str}(X/k)$ with $\mathscr{V}$ in $\text{str}^{\text{fin}}(X/k)$ we also have $\mathscr{W}$ in $\text{str}^{\text{fin}}(X/k)$. Indeed, let $Y \xrightarrow{p} X$ be the G-torsor constructed from $\mathscr{V}$. By assumption, G is an étale finite group scheme. Then $p^*\mathscr{V}$ is a trivial vector bundle on Y equipped with the trivial connection. According to dos Santos' result mentioned above ($\pi^{\text{uni}}(X)$ is a quotient of $\pi^{\text{ét}}(X)$), $p^*\mathscr{W}$ is also a finite connection on Y. But this implies the push forward $p_*p^*\mathscr{W}$ is also a finite connection that forces $\mathscr{W}$ to be a finite connection as we have an inclusion of connections

$$\mathscr{W} \hookrightarrow p_*p^*\mathscr{W}.$$

Thus res^1 is an isomorphism.

For $i \geq 2$, we use the same trick as in the proof of 3.1.1. First notice that the isomorphism res^1 extends to any representations of $\pi^{\text{ét}}(X)$. Let J be the injective envelope of V in $\mathrm{Rep}(\pi^{\text{ét}}(X)) \cong \mathrm{Ind}-\mathscr{C}^{\text{ét}}(X)$. Then we have

$$\mathrm{H}^2(\pi^{\text{ét}}(X), V) \cong \mathrm{H}^1(\pi^{\text{ét}}(X), J/V) \overset{(\mathrm{res}^1)}{\cong} \mathrm{H}^1(\pi(X), J/V) \cong \mathrm{H}^2(\pi(X), V) = 0.$$

The proof is complete. □

As a corollary we obtain the old result due to Serre and Pacheco-Stevenson [Ser90, PS00].

Corollary 3.2.2. *The cohomology groups* $\mathrm{H}^i(\pi^{\text{ét}}(X), V)$ *vanish for* $i \geq 2$.

3.3. **Freeness of the pro-unipotent fundamental group scheme.** By definition, the cohomology of $\pi^{\mathrm{uni}}(X)$ with coefficients in a representation V is the same as the cohomology of $(\mathscr{V}, \nabla)$ in $\mathrm{str}^{\mathrm{uni}}(X/k)$ with respect to its ind-category, where $(\mathscr{V}, \nabla)$ is the nilpotent stratified bundle corresponding to the representation V:

$$\mathrm{H}^i(\pi^{\mathrm{uni}}(X), V) \cong \mathrm{H}^i_{\mathrm{str}^{\mathrm{uni}}}(X/k, (\mathscr{V}, \nabla)).$$

Using Tannakian duality and the fact that extensions of nilpotent stratified bundles in $\mathrm{str}(X/k)$ are again nilpotent, we have, for $i = 0, 1$,

$$\mathrm{H}^i_{\mathrm{str}^{\mathrm{uni}}}(X/k, (\mathscr{V}, \nabla)) \cong \mathrm{H}^i_{\mathrm{str}}(X/k, (\mathscr{V}, \nabla))$$

and

$$\mathrm{H}^i(\pi^{\mathrm{uni}}(X), V) \cong \mathrm{H}^i(\pi(X), V)$$

for any nilpotent stratified bundle $(\mathscr{V}, \nabla)$ and its fiber V. Recall that the latter isomorphism extends to ind-representations, while the former one may not extend to ind-stratified bundles.

Corollary 3.3.1. *The group cohomology* $\mathrm{H}^i(\pi^{\mathrm{uni}}(X), V)$, $i \geq 2$, *vanishes for all representations* V.

Proof. Let J be an injective envelope of V in $\mathrm{Rep}(\pi^{\mathrm{uni}}(X/k))$. Then we have a natural isomorphism (cf. proof of Proposition 3.2.1):

$$\mathrm{H}^2(\pi^{\mathrm{uni}}(X), V) \cong \mathrm{H}^1(\pi^{\mathrm{uni}}(X), J/V).$$

Employing the isomorphism

$$\mathrm{H}^1(\pi^{\mathrm{uni}}(X), V) \cong \mathrm{H}^1(\pi(X), V)$$

which holds for any representation of $\pi^{\mathrm{uni}}(X)$, we obtain an isomorphism

$$\mathrm{H}^2(\pi^{\mathrm{uni}}(X), V) \hookrightarrow \mathrm{H}^2(\pi(X), V).$$

Whence we obtain the vanishing claim, by means of Theorem 3.1.1. The case of $i \geq 3$ follows by a similar argument. □

We now introduce the notion of freeness in the category of pro-étale pro-unipotent group schemes. This is the natural analogue, on the group-scheme side, of the usual universal property of a free pro-p group.

Definition 3.3.2. A pro-étale pro-unipotent affine group scheme over k is an affine group scheme G which can be written as a projective limit

$$G = \varprojlim_i G_i,$$

where each G_i is a finite étale unipotent group scheme over k. Equivalently, since k is algebraically closed, the group schemes G_i are finite constant p-group schemes.

Definition 3.3.3. Let S be a set. A pro-étale pro-unipotent affine group scheme $F_S^{\mathrm{uni,et}}$ over k is called the free pro-étale pro-unipotent group scheme generated by S if, for every pro-étale pro-unipotent affine group scheme G over k, there is a natural bijection

$$\mathrm{Hom}_{k\text{-grp.sch}}\big(F_S^{\mathrm{uni,et}}, G\big) \simeq \mathrm{Map}(S, G(k)).$$

We first establish the equivalence between the category of pro-p groups and the category of pro-étale pro-unipotent affine group schemes over k.

Proposition 3.3.4. *The functor*

$$P \longmapsto \underline{P}_k$$

from pro-p groups to pro-étale pro-unipotent affine group schemes over k is an equivalence of categories. Its quasi-inverse is given by

$$G \longmapsto G(k).$$

Proof. We first prove the assertion at the finite level. Since k is algebraically closed, every finite étale k-scheme is a finite disjoint union of copies of $\mathrm{Spec}\, k$ [Stack, Tag 04GL]. Therefore every finite étale group scheme H over k is constant:

$$H \simeq \underline{H(k)}_k.$$

Thus the category of finite étale group schemes over k is equivalent to the category of finite groups.

Under this equivalence, the representation category of $\underline{\Gamma}_k$ is the category of $k[\Gamma]$-modules. Hence $\underline{\Gamma}_k$ is unipotent if and only if every simple $k[\Gamma]$-module is trivial. If Γ is a finite p-group, then $k[\Gamma]$ is a local algebra; equivalently, it has a unique simple module, namely the trivial one. Conversely, if Γ is not a p-group, choose a prime $\ell \neq p$ dividing $|\Gamma|$. By Cauchy's theorem, Γ contains a cyclic subgroup $C \simeq \mathbb{Z}/\ell\mathbb{Z}$. Since k is algebraically closed and $\ell \neq p$, C has a non-trivial one-dimensional character over k. Inducing this character to Γ and taking a simple quotient gives a non-trivial simple $k[\Gamma]$-module. Hence $\underline{\Gamma}_k$ is unipotent if and only if Γ is a finite p-group.

Now let $P = \varprojlim_i P_i$ be a pro-p group, with P_i finite p-groups. Define

$$\underline{P}_k := \varprojlim_i \underline{P_i}_k.$$

This is a pro-étale pro-unipotent affine group scheme. Conversely, if $G = \varprojlim_i G_i$ is pro-étale pro-unipotent, then each G_i is finite étale unipotent, hence

$$G_i \simeq \underline{G_i(k)}_k$$

with $G_i(k)$ a finite p-group. Therefore

$$G(k) = \varprojlim_i G_i(k)$$

is a pro-p group and

$$G \simeq \varprojlim_i \underline{G_i(k)}_k = \underline{G(k)}_k.$$

It remains to prove that the functor is fully faithful. Write

$$P = \varprojlim_i P_i, \qquad Q = \varprojlim_j Q_j$$

as inverse limits of finite p-groups. Since $\underline{Q}_k = \varprojlim_j \underline{Q_j}_k$, a morphism $\underline{P}_k \to \underline{Q}_k$ is the same thing as a compatible family of morphisms

$$\underline{P}_k \longrightarrow \underline{Q_j}_k.$$

For fixed j, the group scheme $\underline{Q_j}_k$ is finite étale, hence of finite presentation over k. Since $\underline{P}_k = \varprojlim_i \underline{P_i}_k$ is an inverse limit with affine transition morphisms, every morphism $\underline{P}_k \to \underline{Q_j}_k$ factors through some finite stage $\underline{P_i}_k$, by the standard limit argument for morphisms into a finitely presented scheme [Stack, Tag 01ZM]. Hence

$$\begin{aligned}
\mathrm{Hom}_{k\text{-grp.sch}}\big(\underline{P}_k, \underline{Q}_k\big) &\simeq \varprojlim_j \mathrm{Hom}_{k\text{-grp.sch}}\big(\underline{P}_k, \underline{Q_j}_k\big) \\
&\simeq \varprojlim_j \varinjlim_i \mathrm{Hom}_{k\text{-grp.sch}}\big(\underline{P_i}_k, \underline{Q_j}_k\big) \\
&\simeq \varprojlim_j \varinjlim_i \mathrm{Hom}_{\mathrm{grp}}(P_i, Q_j).
\end{aligned}$$

On the other hand, a continuous homomorphism $P \to Q$ is the same thing as a compatible family of continuous homomorphisms $P \to Q_j$. Since Q_j is finite and discrete, each such homomorphism factors through a finite quotient P_i. Thus

$$\mathrm{Hom}_{\mathrm{cont.grp}}(P, Q) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}_{\mathrm{grp}}(P_i, Q_j).$$

This proves full faithfulness, and hence the proposition. □

Lemma 3.3.5. *Let P be a pro-p group and let $\underline{P}_k$ be the associated pro-constant affine group scheme over k. For every discrete continuous k-linear representation V of P, viewed equivalently as a representation of $\underline{P}_k$, there is a natural isomorphism*

$$\mathrm{H}^n(\underline{P}_k, V) \simeq \mathrm{H}^n_{\mathrm{cts}}(P, V).$$

In particular,

$$\mathrm{H}^n(\underline{P}_k, k) \simeq \mathrm{H}^n_{\mathrm{cts}}(P, k).$$

Moreover, if M is a finite discrete $\mathbb{F}_p[P]$-module, then

$$\mathrm{H}^n_{\mathrm{cts}}(P, M) \otimes_{\mathbb{F}_p} k \simeq \mathrm{H}^n_{\mathrm{cts}}\big(P, M \otimes_{\mathbb{F}_p} k\big).$$

Proof. The category $\mathrm{Rep}_k(P_k)$ identifies with the category of discrete continuous k-linear representations of P. Under this identification, the invariant functor is the usual invariant functor

$$V \longmapsto V^P.$$

Therefore its right derived functors agree with continuous group cohomology:

$$\mathrm{H}^n(P_k, V) = R^n(V \mapsto V^P) = \mathrm{H}^n_{\mathrm{cts}}(P, V).$$

Equivalently,

$$\mathrm{H}^n(P_k, V) = \mathrm{Ext}^n_{\mathrm{Rep}_k(P_k)}(k, V) \simeq \mathrm{H}^n_{\mathrm{cts}}(P, V).$$

For the last assertion, compute $\mathrm{H}^n_{\mathrm{cts}}(P, M)$ by the standard complex of continuous inhomogeneous cochains

$$C^q_{\mathrm{cts}}(P, M) = \mathrm{Cont}(P^q, M).$$

Since M is finite and k is flat over $\mathbb{F}_p$, tensoring the continuous cochain complex with k gives the cochain complex computing

$$\mathrm{H}^\bullet_{\mathrm{cts}}(P, M \otimes_{\mathbb{F}_p} k).$$

Hence

$$\mathrm{H}^n_{\mathrm{cts}}(P, M) \otimes_{\mathbb{F}_p} k \simeq \mathrm{H}^n_{\mathrm{cts}}(P, M \otimes_{\mathbb{F}_p} k).$$

□

Proposition 3.3.6. *The group scheme $\pi^{\mathrm{uni}}(X)$ is a free pro-étale pro-unipotent group scheme on $h^1(\pi^{\mathrm{uni}}, k)$ generators.*

Proof. We apply Proposition 3.3.4 to $\pi^{\mathrm{uni}} := \pi^{\mathrm{uni}}(X)$ which is a pro-étale pro-unipotent group scheme (see Section 3.2). Hence there exists a pro-p group P such that

$$\pi^{\mathrm{uni}} \simeq \underline{P}_k.$$

By Lemma 3.3.5, the vanishing

$$\mathrm{H}^2(\pi^{\mathrm{uni}}, W) = 0$$

for every rational representation W of π^{uni} implies, in particular, that

$$\mathrm{H}^2_{\mathrm{cts}}(P, M \otimes_{\mathbb{F}_p} k) = 0$$

for every finite discrete $\mathbb{F}_p[P]$-module M. By the base-change isomorphism in Lemma 3.3.5, this gives

$$\mathrm{H}^2_{\mathrm{cts}}(P, M) \otimes_{\mathbb{F}_p} k = 0.$$

Since $k/\mathbb{F}_p$ is faithfully flat, it follows that

$$\mathrm{H}^2_{\mathrm{cts}}(P, M) = 0$$

for every finite discrete $\mathbb{F}_p[P]$-module M. Hence

$$\mathrm{cd}_p(P) \leq 1.$$

By Serre's criterion for pro-p groups [Ser02, Chapter I, Section 4], a pro-p group has p-cohomological dimension at most 1 if and only if it is free pro-p. Therefore P is free pro-p.

Let

$$r := \dim_k \mathrm{H}^1(\pi^{\mathrm{uni}}, k).$$

Then

$$\begin{aligned} r &= \dim_k \mathrm{H}^1(\pi^{\mathrm{uni}}, k) \\ &= \dim_k \big(\mathrm{H}^1_{\mathrm{cts}}(P, \mathbb{F}_p) \otimes_{\mathbb{F}_p} k\big) \\ &= \dim_{\mathbb{F}_p} \mathrm{H}^1_{\mathrm{cts}}(P, \mathbb{F}_p). \end{aligned}$$

For a free pro-p group, this number is precisely the minimal number of topological generators. Hence

$$P \simeq F_r^{(p)},$$

where $F_r^{(p)}$ denotes the free pro-p group on r generators. Consequently

$$\pi^{\mathrm{uni}} \simeq \underline{F_r^{(p)}}_k.$$

It remains to express this conclusion intrinsically in the language of group schemes (see Definition 3.3.3). Let S be a set with $|S| = r$. For every pro-étale pro-unipotent affine group scheme G over k, write $G \simeq \underline{Q}_k$, where $Q = G(k)$. Since $P \mapsto P_k$ is an equivalence of categories, we get

$$\begin{aligned}\mathrm{Hom}_{k\text{-grp.sch}}(\pi^{\mathrm{uni}}, G) &\simeq \mathrm{Hom}_{k\text{-grp.sch}}\big(\underline{F_r^{(p)}}_k, \underline{Q}_k\big)\\ &\simeq \mathrm{Hom}_{\mathrm{cont.grp}}\big(F_r^{(p)}, Q\big)\\ &\simeq \mathrm{Map}(S, Q)\\ &= \mathrm{Map}(S, G(k)).\end{aligned}$$

Thus π^{uni} is the free pro-étale pro-unipotent group scheme on $r = h^1(\pi^{\mathrm{uni}}, k)$ generators. □

3.4. **Application: a computation for cohomology of curves of genus** 1**.** Let X have genus 1. Then, according to dos Santos [dSa07], $\pi(X)$ is commutative and decomposes into a direct product of a unipotent part and a diagonal part. Since the group cohomology of the diagonal part vanishes (being linearly reductive),

$$\mathrm{H}^i(\pi(X), V) = \mathrm{H}^i(\pi^{\mathrm{uni}}(X), V),$$

for $i \geq 1$ and any unipotent representation V. In categorical terms, we have:

- If X is supersingular, the p-rank is zero, so the unipotent part of $\pi(X)$ is trivial. Hence $\pi(X)$ is diagonal.
- If X is ordinary, that is, the absolute Frobenius acts by a bijective map, then according to [dSa07], any object of $\mathrm{str}^{\mathrm{coh}}(X/k)$ decomposes into a direct sum of objects of the form

  $$\mathscr{L} \otimes \mathscr{N}$$

  where $\mathscr{L}$ is a rank one stratified bundle and $\mathscr{N}$ is a nilpotent object, i.e., all of its irreducible subquotients are trivial objects. Such an object is also called $\mathscr{L}$-isotypical.

Now, dos Santos tells us that (loc.cit. Theorem 21)

$$\pi^{\mathrm{uni}}(X) = \mathbb{Z}_p^r,$$

where r is the p-rank of X (i.e., the rank of the Frobenius map acting on $\mathrm{H}^1(X, \mathscr{O}_X)$, which can be either 1 or 0). Thus we have a complete understanding of the stratified cohomology of stratified bundles on an elliptic curve.

Proposition 3.4.1. *Let X be a smooth projective curve of genus 1.*

(1) *If X has p-rank 0 then $\mathrm{h}^1_{\mathrm{str}}(X, \mathscr{E}) = 0$ for any stratified bundle $\mathscr{E}$.*
(2) *If X has p-rank 1, then $\mathrm{h}^1_{\mathrm{str}}(X, \mathscr{O}_X) = 1$ and hence this function is non-zero at any unipotent stratified bundle. On the other hand, for any non-trivial stratified line bundle $\mathscr{L}$, we have $\mathrm{h}^1_{\mathrm{str}}(X, \mathscr{V}) = 0$ if $\mathscr{V}$ is an $\mathscr{L}$-isotypical stratified bundle.*

Proof. If X has p-rank 0, that is, X is supersingular, the action of the Frobenius of X is zero, hence $\pi(X)$ is diagonal. So its cohomology vanishes from degree 1.

If X has p-rank 1, that is X is ordinary, the absolute Frobenius acts by a bijective map on the 1-dimensional space $\mathrm{H}^1(X,\mathscr{O}_X)$. Hence $\mathrm{H}^1_{\mathrm{str}}(X,\mathscr{O}_X)\cong k$. If $\mathscr{V}$ is a nilpotent object, then there is a surjective map $\mathscr{V}\longrightarrow\mathscr{O}_X$ as objects in $\mathrm{str}(X/k)$. As the $\mathrm{H}^2_{\mathrm{str}}$-group vanishes, we have a surjective map $\mathrm{H}^1_{\mathrm{str}}(X,\mathscr{V})\longrightarrow\mathrm{H}^1_{\mathrm{str}}(X,\mathscr{O}_X)$, whence $\mathrm{H}^1_{\mathrm{str}}(X,\mathscr{V})\neq 0$.

On the other hand, as explained above, any object of $\mathrm{str}^{\mathrm{coh}}(X/k)$ decomposes into a direct sum of objects of the form $\mathscr{L}\otimes\mathscr{N}$ where $\mathscr{L}$ is a rank one stratified bundle and $\mathscr{N}$ is a nilpotent object, i.e., all of its irreducible subquotients are trivial. Hence, if $\mathscr{L}$ is a non-trivial rank 1 object $\mathrm{H}^1_{\mathrm{str}}(X,\mathscr{L})\cong\mathrm{Ext}^1_{\mathrm{str}(X/k)}(\mathscr{O}_X,\mathscr{L})=0$. Using the long exact sequence of cohomology groups, we obtain $\mathrm{Ext}^1_{\mathrm{str}(X/k)}(\mathscr{N},\mathscr{L})=0$ for any nilpotent stratified bundle $\mathscr{N}$, whence the last claim of Proposition. □

Corollary 3.4.2. *Consider $\mathbb{Z}_p$ as a pro-finite group scheme over k. Then $\mathrm{h}^i(\mathbb{Z}_p,-)$ vanishes for any $i\geq 2$ and*

$$\mathrm{h}^i(\mathbb{Z}_p,k)=1,$$

for $i=0,1$.

3.5. **The non-vanishing of the first cohomology group.** Our aim in this section is to discuss the vanishing property of the first cohomology group. This question is motivated by the non-vanishing property for the first de Rham cohomology on curves over a field of characteristic zero [BHT25]. We shall restrict ourselves to *finite stratified bundles*, that is étale finite (i.e. trivializable) vector bundles, cf. 3.2.

A vector bundle is said to be F-periodic if it is isomorphic to the pull-back of itself by some powers of the Frobenius map. According to Lange-Stuhler [LS77], F-periodic vector bundles are étale trivializable. The converse implication holds if the base field is finite or if the vector bundle is stable [BD07]. Now we restrict further our problem to F-periodic vector bundles.

Let $\mathscr{E}$ be an F-periodic vector bundle, we shall use the same symbol for the associated stratified bundle. Fix an isomorphism $\tau:F^{*n}\mathscr{E}\longrightarrow\mathscr{E}$. The composition of τ's and the canonical inclusion $\mathscr{E}\longrightarrow F^{*n}\mathscr{E}$ yields a p^n-linear endomorphism

$$f:\mathscr{E}\longrightarrow F^{*n}\mathscr{E}\xrightarrow{\tau}\mathscr{E}.$$

Denote by $\mathscr{H}_{\mathscr{E}}$ the subsheaf of $\mathscr{E}$ fixed by f in the étale topology on X. The lemma below goes back to [Kat69, Propositions 1.1 and 1.2].

Lemma 3.5.1. *Let $\mathscr{E}$ be an F-periodic stratified bundle of period n. Then $\mathscr{H}_{\mathscr{E}}$ is a locally constant sheaf with fiber a vector space over $\mathbb{F}_{p^n}$ with dimension equal to the rank of $\mathscr{E}$, and there is an exact sequence*

$$0\to\mathscr{H}_{\mathscr{E}}\longrightarrow\mathscr{E}\xrightarrow{f-1}\mathscr{E}\longrightarrow 0$$

of sheaves on $X_{\acute{e}t}$.

Remark. $\mathscr{H}_{\mathscr{E}}$ can be seen as a finite additive group scheme on X.

Proposition 3.5.2. *Let $\mathscr{E}$ be an F-periodic stratified bundle of period n, and let $\mathscr{H}_{\mathscr{E}}$ be the associated finite étale group scheme defined above. Then we have a canonical isomorphism:*

$$\mathrm{H}^i_{\acute{e}t}(X, \mathscr{H}_{\mathscr{E}}) \otimes_{\mathbb{F}_q} k \xrightarrow{\cong} \mathrm{H}^i_{\mathrm{str}}(X, \mathscr{E}),$$

for all i, where $q = p^n$.

Proof. From the short exact sequence of sheaves on $X_{\text{ét}}$:

$$0 \to \mathscr{H}_{\mathscr{E}} \to \mathscr{E} \xrightarrow{F^{*n}-1} \mathscr{E} \to 0,$$

we have the following long exact sequence of *étale* cohomologies:

$$\cdots \to \mathrm{H}^i_{\text{ét}}(X, \mathscr{H}_{\mathscr{E}}) \to \mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \xrightarrow{F^{*n}-1} \mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \to \cdots \tag{5}$$

Since $\mathscr{E}$ is a coherent sheaf on X, we have $\mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \cong \mathrm{H}^i(X, \mathscr{E})$. Since the induced morphism on cohomology $F^n : \mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \to \mathrm{H}^i_{\text{ét}}(X, \mathscr{E})$ is q-linear, the argument before this theorem implies that $F^{*n} - 1 : \mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \to \mathrm{H}^i_{\text{ét}}(X, \mathscr{E})$ is surjective (cf. [Kat69, Proposition 1.2]). Hence, for each i, we have the following short exact sequence:

$$0 \to \mathrm{H}^i_{\text{ét}}(X, \mathscr{H}_{\mathscr{E}}) \to \mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \xrightarrow{F^{*n}-1} \mathrm{H}^i_{\text{ét}}(X, \mathscr{E}) \to 0.$$

This finishes the proof since we have $\mathrm{H}^i_{\mathrm{str}}(X, \mathscr{E}) = \mathrm{H}^i(X, \mathscr{E})_{\mathrm{ss}} \cong \mathrm{H}^i(X, \mathscr{E})^{F^{*n}-1} \otimes_{\mathbb{F}_q} k$ which is isomorphic to $\mathrm{H}^i_{\text{ét}}(X, \mathscr{H}_{\mathscr{E}}) \otimes_{\mathbb{F}_q} k$ by the above short exact sequence. □

Assume that X is a curve of genus larger than one. The method of [BHT25] relies on the Euler characteristic formula which is not always available in positive characteristic. More precisely, let $\mathscr{F}$ be a locally constant sheaf on X and $f : Y \longrightarrow X$ be an étale covering map that trivializes $\mathscr{F}$, that is, $f^*\mathscr{F}$ is a constant sheaf on Y. Then it holds [Pin00, Theorem 0.2]:

$$\chi(X, \mathscr{F}) \geq (1 - g_X) \cdot \operatorname{rank} \mathscr{F},$$

where $\operatorname{rank} \mathscr{F}$ denotes the $\mathbb{F}_p$-dimension of the fiber of $\mathscr{F}$ at a point of X. The equality takes place when Y is ordinary. We introduce the following notation.

Definition 3.5.3. A vector bundle $\mathscr{F}$ on a smooth projective curve X is said to be ordinarily étale trivializable if there exists a Galois étale covering map $f : Y \longrightarrow X$ that trivializes $\mathscr{F}$, where Y is an ordinary curve.

Corollary 3.5.4. *Let X be an ordinary curve of genus larger than* 1*. Let $\mathscr{E}$ be an ordinarily étale trivializable vector bundle. Then* $\mathrm{h}^1_{\mathrm{str}}(X, \mathscr{E}) \neq 0$.

Proof. Étale trivializable vector bundles are semi-stable of degree 0, hence possess composition series such that the successive quotients are stable of degree zero. Using the long exact sequence of cohomology and induction on the length of the above composition series, the problem is essentially reduced to considering only étale trivializable *stable* vector bundles. According to [BD07], étale trivializable stable vector bundles are F-periodic. Thus, we can assume that $\mathscr{E}$ is F-periodic with F-period, say, n.

By Proposition 3.5.2, it is enough to show that $\mathrm{h}^1_{\text{ét}}(X, \mathscr{H}_{\mathscr{E}}) \neq 0$. By [Pin00, Theorem 0.2], we have that:

$$\chi(H_{\mathscr{E}}) = (1 - \mathrm{h}^1_{\mathrm{str}}(X, \mathscr{O}_X)) \cdot \operatorname{rank} \mathscr{H}_{\mathscr{E}} \neq 0.$$

This proves our claim. □

We shall give in the Appendix A.6 an example of an F-periodic vector bundle whose first stratified cohomology vanishes.

4. Comparison for cohomologies of ind-stratified bundles

For any affine group scheme G over k, rational group cohomology commutes with filtered direct limits:

$$\mathrm{H}^i\big(G, \varinjlim_\alpha V_\alpha\big) \cong \varinjlim_\alpha \mathrm{H}^i(G, V_\alpha).$$

Indeed, the standard Hochschild complex computing this cohomology has terms $C^n(G,V) = V \otimes_k k[G]^{\otimes n}$ (cf. [Jan87, Part I, Chapter 4]). These commute with filtered direct limits, which are exact over k, so taking cohomology commutes with these limits as well. In contrast, stratified cohomology involves inverse limits and need not commute with filtered direct limits. Consequently, the comparison maps of Theorem 3.1.1 need not be isomorphisms for ind-stratified bundles, as the examples below show.

4.1. **The case of elliptic curves.** We first discuss the comparison problem for ind-stratified bundles on elliptic curves. The answer depends on whether the curve is supersingular or ordinary.

Proposition 4.1.1. *Let X/k be an elliptic curve.*

(1) *If X is supersingular, then the maps δ^i defined as in Theorem 3.1.1 are isomorphisms for every ind-stratified bundle and every $i \geq 0$.*
(2) *If X is ordinary, then the same conclusion holds for every $\mathscr{L}$-isotypical ind-stratified bundle, where $\mathscr{L}$ is a rank-one stratified bundle. Consequently, it also holds for finite direct sums of isotypical ind-stratified bundles.*

Proof. Assume first that X is supersingular. By [dSa07, Theorem 21], the unipotent part of $\pi(X)$ is trivial, hence $\pi(X)$ is diagonal. Thus every ind-stratified bundle is a direct sum of rank-one objects.

Let $\mathscr{L}$ be a non-trivial stratified line bundle. Since $\mathrm{Pic}^0(X)[p](k) = 0$, multiplication by p is injective on $\mathrm{Pic}^0(X)(k)$. It follows that $\mathscr{L}^{(n)}$ is non-trivial for every n: indeed, if one term were trivial, then all terms would be trivial and hence so would $\mathscr{L}$. Therefore

$$\mathrm{H}^0(X, \mathscr{L}^{(n)}) = \mathrm{H}^1(X, \mathscr{L}^{(n)}) = 0 \qquad \text{for all } n.$$

For the trivial character, the transition maps on $\mathrm{H}^0(X, \mathscr{O}_X)$ are bijective, whereas those on $\mathrm{H}^1(X, \mathscr{O}_X)$ are zero. Ogus' exact sequence [Ogu75, Theorem 2.4] therefore gives

$$\mathrm{H}^i_{\mathrm{str}}(X/k, \mathscr{V}) = 0 \qquad (i \geq 1)$$

for every ind-stratified bundle $\mathscr{V}$. The same vanishing holds for group cohomology because $\pi(X)$ is diagonal.

Now assume that X is ordinary and let $\mathscr{V} = \mathscr{L} \otimes \mathscr{N}$ be $\mathscr{L}$-isotypical, with $\mathscr{N}$ ind-unipotent. If $\mathscr{L}$ is trivial, Frobenius acts bijectively on $\mathrm{H}^i(X, \mathscr{O}_X)$ for $i = 0, 1$. By induction on the length of

a finite unipotent object, and then by passage to filtered direct limits, all transition maps

$$\mathrm{H}^i(X,\mathscr{N}^{(n+1)}) \longrightarrow \mathrm{H}^i(X,\mathscr{N}^{(n)}), \qquad i=0,1,$$

are bijective. Hence the first derived inverse limits vanish and

$$\mathrm{H}^1_{\mathrm{str}}(X/k,\mathscr{N}) \cong \mathrm{H}^1(X,\mathscr{N}) \cong \varinjlim_{\alpha} \mathrm{H}^1_{\mathrm{str}}(X/k,\mathscr{N}_\alpha),$$

where $(\mathscr{N}_\alpha)$ is the system of coherent unipotent subobjects. Since group cohomology commutes with filtered direct limits, Theorem 3.1.1 gives the comparison in this case.

If $\mathscr{L}$ is non-trivial, there exists n_0 such that $\mathscr{L}^{(n)}$ is a non-trivial degree-zero line bundle for every $n \geq n_0$. A non-trivial degree-zero line bundle on an elliptic curve has vanishing cohomology. Using finite unipotent filtrations and then filtered direct limits, we obtain

$$\mathrm{H}^i\big(X,\mathscr{L}^{(n)}\otimes\mathscr{N}^{(n)}\big)=0, \qquad i=0,1, \quad n\geq n_0.$$

Thus the inverse systems occurring in Ogus' exact sequence [Ogu75, Theorem 2.4] are eventually zero. The group-cohomology side also vanishes, by Theorem 3.1.1 on coherent subobjects and commutation of group cohomology with filtered direct limits. This proves the assertion for an isotypical ind-object, and the statement for finite direct sums follows. □

The restriction to finite direct sums in Proposition 4.1.1(2) cannot be removed in general.

Example 4.1.2 (suggested by ChatGPT)**.** Let X/k be an ordinary elliptic curve. Choose $0 \neq Q_0 \in \mathrm{Pic}^0(X)[p](k)$ and recursively $Q_{m+1} \in \mathrm{Pic}^0(X)(k)$ with $Q_{m+1}^{\otimes p} \simeq Q_m$; this is possible since multiplication by p on $\mathrm{Pic}^0(X)(k)$ is surjective. All Q_m are non-trivial. For $j \geq 1$, define

$$\mathscr{L}_j^{(n)} = \begin{cases} \mathscr{O}_X, & n<j, \\ Q_{n-j}, & n\geq j. \end{cases}$$

Since $F^*Q \simeq Q^{\otimes p}$, together with $Q_0^{\otimes p} \simeq \mathscr{O}_X$ and $Q_{m+1}^{\otimes p} \simeq Q_m$, these terms carry the required Frobenius transition isomorphisms and hence define a stratified line bundle $\mathscr{L}_j$. Set $\mathscr{V}_N = \bigoplus_{j=1}^N \mathscr{L}_j$ and $\mathscr{V} = \bigoplus_{j\geq 1}\mathscr{L}_j$. For $n \geq N$, all summands of $\mathscr{V}_N^{(n)}$ are non-trivial line bundles of degree zero, hence

$$\mathrm{H}^q(X,\mathscr{V}_N^{(n)})=0, \qquad q=0,1.$$

Therefore Ogus' exact sequence [Ogu75, Theorem 2.4] gives

$$\mathrm{H}^1_{\mathrm{str}}(X/k,\mathscr{V}_N)=0 \qquad \text{for all } N.$$

For the infinite sum,

$$A_n := \mathrm{H}^0(X,\mathscr{V}^{(n)}) = \bigoplus_{j>n} k e_j,$$

where e_j is the unit section of the j-th trivial summand. Regard (A_n,φ_n) as an inverse system of abelian groups; the transition $\varphi_n : A_{n+1} \to A_n$ is Frobenius on the common coordinates. The sequence $y_n = e_{n+1}$ defines a non-zero class in $\varprojlim_n{}^1 A_n$: if $y_n = x_n - \varphi_n(x_{n+1})$, then for every m

$$x_0 = e_1 + \cdots + e_m + \varphi_{0,m}(x_m),$$

which is impossible since $x_0 \in A_0 = \bigoplus_{j\geq 1} ke_j$ has finite support. Thus $\varprojlim_n^1 A_n \neq 0$, and [Ogu75, Theorem 2.4] gives $\mathrm{H}^1_{\mathrm{str}}(X/k, \mathscr{V}) \neq 0$. On the other hand, if $V_N = \mathscr{V}_N|_x$ and $V = \mathscr{V}|_x$, then Theorem 3.1.1 and the commutation of group-scheme cohomology with filtered direct limits give

$$\mathrm{H}^1(\pi(X), V) = \varinjlim_N \mathrm{H}^1(\pi(X), V_N) = 0.$$

Hence δ^1 is not bijective for $\mathscr{V}$.

Remark 4.1.3. The point of the preceding example is that, although each non-trivial rank-one component eventually has vanishing cohomology, there is no uniform Frobenius level from which this happens simultaneously for all components. For every finite direct sum such a uniform level exists, which explains why Proposition 4.1.1(2) still applies in that case. For the infinite direct sum above, the unbounded Frobenius delay produces a non-zero $\varprojlim^1$ term, and hence a stratified cohomology class which is not detected at any finite stage.

This phenomenon is different from the counterexample constructed below for curves of genus 2. There the ind-stratified bundle lies entirely in the unipotent isotypical component, and the failure of comparison is caused instead by an unbounded order of Cartier nilpotency.

4.2. **The order of Cartier nilpotency of a Frobenius invariant vector bundle.** Next we show that the comparison map δ^i for curves of genus at least 2 may be non-bijective. For a counterexample, we construct an infinitely iterated extension of the trivial vector bundle which is Frobenius invariant and investigate its stratified cohomology. For this, we introduce the concept of the order of Cartier nilpotency of a Frobenius invariant vector bundle, which seems interesting on its own.

Our approach is based on the Cartier map. Let Ω_X denote the sheaf of differential forms on X. Let $\mathscr{B}_X$ denote the image of $F_{X*}\mathscr{O}_X$ in $F_{X*}\Omega_X$ under the differential $F_{X*}d$. This is called the sheaf of locally exact forms. Then we have exact sequences

$$0 \longrightarrow \mathscr{O}_X \longrightarrow F_{X*}\mathscr{O}_X \longrightarrow \mathscr{B}_X \longrightarrow 0$$

and

$$0 \longrightarrow \mathscr{B}_X \longrightarrow F_{X*}\Omega_X \xrightarrow{C} \Omega_X \longrightarrow 0,$$

where C denotes the Cartier map [Car57]. Roughly speaking, the differential forms that are not locally exact have the form $u^{p-1}du$ where u is not a p-power (in other words, K is a separable extension of $k(u)$). Such a form is mapped to du by the Cartier map C, other forms are mapped to zero by C. Serre's duality yields a perfect pairing given by the residue map (cf. [Ser58]):

$$\mathrm{H}^1(X, \mathscr{O}_X) \times \Omega(X) \longrightarrow k, \quad (e, \mu) \longmapsto \langle e, \mu\rangle = \mathrm{Res}(e_{ij}\mu) := \sum_{P\in|X|} \mathrm{res}_P(f_i\mu).$$

Then C is conjugate to F_X^* with respect to this pairing in the sense that $\langle F_X^* e, \mu\rangle = \langle e, C\mu\rangle^p$. In fact, we have, for any closed point P, $\mathrm{res}_P(\mu) = \mathrm{res}_P(C\mu)^p$.

Consider the restriction of C on the space $\Omega(X)$ of global 1-forms. C is p^{-1}-linear. Hence $\Omega(X)$ decomposes into the direct sum of a "C-bijective" subspace and a "C-nilpotent" subspace. This decomposition is compatible with the decomposition of $\mathrm{H}^1(X, \mathscr{O}_X)$ given above.

Let $\mathscr{E}$ be a vector bundle. Tensoring the second sequence with $\mathscr{E}$ and taking the cohomology we get

$$\mathrm{H}^0(X, \mathscr{E} \otimes F_{X*}\Omega_X) \longrightarrow \mathrm{H}^0(X, \mathscr{E} \otimes \Omega_X).$$

By the projection formula

$$\mathscr{E} \otimes F_{X*}\Omega_X \cong F_{X*}(F^*\mathscr{E} \otimes \Omega_X)$$

and, as F_X is affine,

$$\mathrm{H}^0(X, F_{X*}\mathscr{H}) \cong \mathrm{H}^0(X, \mathscr{H})$$

for any quasi-coherent sheaf on X ([Har77, Ex. 8.1]), hence we obtain a map

$$C_{\mathscr{E}} : \mathrm{H}^0(X, F^*\mathscr{E} \otimes \Omega_X) \longrightarrow \mathrm{H}^0(X, \mathscr{E} \otimes \Omega_X),$$

called *the Cartier map* for $\mathscr{E}$. This map is conjugate to the Frobenius map on $\mathrm{H}^1(X, \mathscr{E}^\vee) \longrightarrow \mathrm{H}^1(X, F^*\mathscr{E}^\vee)$ by Serre's duality.

Let now $\mathscr{E}$ be Frobenius invariant. Fix an isomorphism $\sigma : \mathscr{E} \longrightarrow F^*\mathscr{E}$. We obtain a p^{-1}-linear map

$$C_\sigma : \mathrm{H}^0(X, \Omega_X \otimes \mathscr{E}) \xrightarrow{\sigma} \mathrm{H}^0(X, \Omega_X \otimes F^*\mathscr{E}) \xrightarrow{C} \mathrm{H}^0(X, \Omega_X \otimes \mathscr{E}). \tag{6}$$

Definition 4.2.1. The order of Cartier nilpotency of the Frobenius-equivariant bundle $(\mathscr{E}, \sigma)$ is the smallest integer n such that $C_\sigma^n = 0$ on the nilpotent part $\mathrm{H}^0(X, \Omega_X \otimes \mathscr{E})$.

The following result will provide a more comfortable way to compute the order of Cartier nilpotency.

Proposition 4.2.2. *Let $C^{n,0}$ denote the composition of n Cartier morphisms:*

$$C_i := C_{F^{*(i-1)}\mathscr{E}} : \mathrm{H}^0(X, \Omega_X \otimes F^{*i}\mathscr{E}) \to \mathrm{H}^0(X, \Omega_X \otimes F^{*(i-1)}\mathscr{E}),$$

$$C^{n,0} := C_1 \circ C_2 \circ \ldots \circ C_n : \mathrm{H}^0(X, \Omega_X \otimes F^{*n}\mathscr{E}) \to \mathrm{H}^0(X, \Omega_X \otimes \mathscr{E}).$$

The order of Cartier nilpotency of $(\mathscr{E}, \sigma)$ is equal to the smallest integer n satisfying that $C^{n,0}$ vanishes on the nilpotent part of $\mathrm{H}^0(X, \Omega_X \otimes F^{*n}\mathscr{E})$.

Proof. The construction of the Cartier map $C_{\mathscr{E}}$ is functorial. In particular, the functoriality with respect to the map $F^{*(i-1)}\sigma : F^{*(i-1)}\mathscr{E} \longrightarrow F^{*i}\mathscr{E}$ amounts to the following commutative diagram:

$$\begin{array}{ccc} \mathrm{H}^0(X, \Omega_X \otimes F^{*i}\mathscr{E}) & \xrightarrow{C_i} & \mathrm{H}^0(X, \Omega_X \otimes F^{*(i-1)}\mathscr{E}) \\ \downarrow{\scriptstyle F^{*i}\sigma} & & \downarrow{\scriptstyle F^{*(i-1)}\sigma} \\ \mathrm{H}^0(X, \Omega_X \otimes F^{*(i+1)}\mathscr{E}) & \xrightarrow{C_{i+1}} & \mathrm{H}^0(X, \Omega_X \otimes F^{*i}\mathscr{E}), \end{array} \tag{7}$$

for any i. Hence, one can check easily by induction that

$$(C \circ \sigma)^n = C_1 \circ C_2 \circ \ldots \circ C_n \circ F^{*(n-1)}\sigma \circ \ldots \circ \sigma.$$

The claim follows. □

4.3. **The construction of the example.** Throughout this subsection, X is a smooth projective curve over the algebraically closed field k of characteristic $p > 0$, with

$$g(X) = 2, \qquad \dim_k \mathrm{H}^1(X, \mathscr{O}_X)_{\mathrm{ss}} = 1. \tag{A}$$

Thus X has p-rank one; equivalently, the stable rank of Frobenius on $\mathrm{H}^1(X, \mathscr{O}_X)$ is one. The Frobenius–Cartier adjunction under Serre duality identifies the duals of the semisimple and

nilpotent summands of $\mathrm{H}^1(X,\mathscr{O}_X)$ with the corresponding summands of $\mathrm{H}^0(X,\Omega_X)$. Both have dimension one, so we fix

$$\mathrm{H}^0(X,\Omega_X) = k\eta \oplus k\omega, \qquad C(\eta) = \eta, \qquad C(\omega) = 0,$$

where $\eta \neq 0$ is chosen in the semisimple part and $0 \neq \omega \in \mathrm{H}^0(X,\Omega_X)_{\mathrm{nil}}$. The normalization of η is possible since k is algebraically closed.

Our aim is to construct an ind-stratified bundle for which the comparison map δ^1 is not bijective. We first choose

$$0 \neq e_2 \in \mathrm{H}^1(X,\mathscr{O}_X)_{\mathrm{ss}}, \qquad F^* e_2 = e_2.$$

Such a class is unique up to multiplication by an element of $\mathbb{F}_p^\times$; see [Mu70, §14, Corollary, p. 143]. The Artin–Schreier sequence identifies the Frobenius-fixed classes in $\mathrm{H}^1(X,\mathscr{O}_X)$ with $\mathrm{H}^1_{\mathrm{et}}(X,\mathbb{F}_p)$. Since the p-rank of X is one, the maximal pro-p quotient of $\pi_1^{\mathrm{et}}(X)$ is $\mathbb{Z}_p$ [Cre84, Theorem 1.9]. We therefore choose a quotient

$$q : \pi_1^{\mathrm{et}}(X) \twoheadrightarrow \mathbb{Z}_p$$

whose reduction modulo p corresponds to e_2.

For $r \geq 1$, put $V_r = \mathbb{F}_p[t]/(t^r)$ and let a topological generator of $\mathbb{Z}_p$ act by multiplication by $1+t$. This action is continuous, since it factors through a finite cyclic p-group. Let $(\mathscr{E}_r,\sigma_r)$ be the corresponding finite stratified bundle, where

$$\sigma_r : \mathscr{E}_r \xrightarrow{\sim} F^*\mathscr{E}_r$$

is the compatible Frobenius isomorphism supplied by this chosen $\mathbb{F}_p$-local system. In particular, $\mathscr{E}_1 = \mathscr{O}_X$, and the extension defining $\mathscr{E}_2$ has class e_2.

We fix once and for all these compatible Frobenius isomorphisms $\sigma_r : \mathscr{E}_r \xrightarrow{\sim} F^*\mathscr{E}_r$; the Cartier operators below are defined relative to this fixed system.

The exact sequence $0 \to V_r \xrightarrow{t} V_{r+1} \to \mathbb{F}_p \to 0$ becomes the extension defining $\mathscr{E}_{r+1}$. Its pushout along $V_r \to V_1$ is the extension defining $\mathscr{E}_2$, whose underlying vector-bundle class is e_2; hence every extension in the tower is nonsplit as a vector bundle. The associated cohomology sequences consequently give

$$\mathrm{H}^0(X,\mathscr{E}_r) = k, \qquad \dim_k \mathrm{H}^1(X,\mathscr{E}_r)_{\mathrm{ss}} = 1,$$

and the maps

$$\mathrm{H}^1(X,\mathscr{E}_r)_{\mathrm{ss}} \longrightarrow \mathrm{H}^1(X,\mathscr{E}_{r+1})_{\mathrm{ss}}$$

are zero. Thus $\mathscr{E} := \varinjlim_r \mathscr{E}_r$ is the desired candidate; the remaining point is to impose the Cartier condition that makes its stratified cohomology large.

The following exact sequences are now part of the construction.

Proposition 4.3.1. *For any positive integers m and n, there is a Frobenius-equivariant exact sequence of vector bundles over X:*

$$0 \longrightarrow \mathscr{E}_n \longrightarrow \mathscr{E}_{m+n} \longrightarrow \mathscr{E}_m \longrightarrow 0.$$

Proof. On the $\mathbb{Z}_p$-modules used above, multiplication by t^m and reduction modulo t^m give an exact sequence

$$0 \longrightarrow \mathbb{F}_p[t]/(t^n) \xrightarrow{t^m} \mathbb{F}_p[t]/(t^{m+n}) \longrightarrow \mathbb{F}_p[t]/(t^m) \longrightarrow 0.$$

Both maps commute with multiplication by $1+t$. Applying the exact functor from these finite representations to finite stratified bundles gives the asserted Frobenius-equivariant sequence. □

For $m, n \geq 1$, write the exact sequence of Proposition 4.3.1 as

$$0 \longrightarrow \mathscr{E}_n \xrightarrow{i_{n,m+n}} \mathscr{E}_{m+n} \xrightarrow{\mathrm{pr}_{m+n,m}} \mathscr{E}_m \longrightarrow 0,$$

and use the same notation for the induced maps on differential forms. The Frobenius equivariance proved in Proposition 4.3.1 says explicitly that

$$F^* i_{n,m+n} \circ \sigma_n = \sigma_{m+n} \circ i_{n,m+n}, \qquad F^* \mathrm{pr}_{m+n,m} \circ \sigma_{m+n} = \sigma_m \circ \mathrm{pr}_{m+n,m}.$$

Set

$$M_r := \mathrm{H}^0(X, \Omega_X \otimes \mathscr{E}_r)_{\mathrm{nil}} \quad (r \geq 1), \qquad M_0 := 0.$$

Denote the resulting Cartier operator on M_r by C_r. In particular, $M_1 = k\omega$; all inverse images below are taken inside M_n. We isolate the additional hypothesis used in the counterexample:

(C) $$C_2(M_2) = 0, \qquad C_3(M_3) \subset i_{1,3}(M_1), \qquad \overline{C}_3 : M_3 / i_{2,3}(M_2) \xrightarrow{\sim} M_1.$$

Here $\overline{C}_3$ is the p^{-1}-linear map induced by C_3, with $i_{1,3}(M_1)$ identified with M_1. Naturality gives $C_3 \circ i_{2,3} = i_{2,3} \circ C_2 = 0$, so C_3 indeed factors through $M_3 / i_{2,3}(M_2)$. Equivalently, in addition to $C_2(M_2) = 0$, a nilpotent lift s_3 of ω satisfies $C_3(s_3) = a\, i_{1,3}(\omega)$ for some $a \neq 0$; the same relation then holds for every such lift. Condition (C) implies Cartier nilpotency order two on M_3. It will be verified in characteristic three in Section 4.4.

Lemma 4.3.2.

i) *For all $m, n \geq 1$, there is an exact sequence*

$$0 \longrightarrow M_n \xrightarrow{i_{n,n+m}} M_{n+m} \xrightarrow{\mathrm{pr}_{n+m,m}} M_m \longrightarrow 0.$$

In particular, $\dim M_n = n$*, and every M_n contains a lift of ω under* $\mathrm{pr}_{n,1}$ *(with* $\mathrm{pr}_{1,1} = \mathrm{id}$*).*

ii) *Assume* (C)*. For $n > 2$ and $s \in M_n$ with* $\mathrm{pr}_{n,1}(s) = \omega$*, there exist $s' \in M_{n-2}$ and $a \in k^\times$ such that*

$$\mathrm{pr}_{n-2,1}(s') = \omega, \qquad C_n(s) = a\, i_{n-2,n}(s').$$

Proof. For i), tensor the sequence in Proposition 4.3.1 with Ω_X and take cohomology. The connecting morphism is Cartier-equivariant. By Serre duality, the Cartier operator on $\mathrm{H}^1(X, \Omega_X \otimes \mathscr{E}_n)$ is dual to Frobenius on $\mathrm{H}^0(X, \mathscr{E}_n^\vee) \simeq k$, hence is bijective. Thus the target of the connecting morphism has trivial nilpotent part. Since the semisimple–nilpotent decomposition of a finite-dimensional vector space with semilinear endomorphism is functorial for equivariant maps, passing to nilpotent parts gives the asserted short exact sequence. Its dimension statement follows from $M_1 = k\omega$.

For ii), put $s_3 = \mathrm{pr}_{n,3}(s)$. The projection $\mathrm{pr}_{n,3} : \mathscr{E}_n \to \mathscr{E}_3$ is compatible with the chosen Frobenius isomorphisms. Naturality of the Cartier map for this bundle morphism therefore

gives the commutative diagram

$$\begin{array}{ccc} M_n & \xrightarrow{C_n} & M_n \\ {\scriptstyle \mathrm{pr}_{n,3}}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_{n,3}} \\ M_3 & \xrightarrow[C_3]{} & M_3. \end{array}$$

Thus condition (C) gives, concretely,

$$\mathrm{pr}_{n,3}\big(C_n(s)\big) = C_3\big(\mathrm{pr}_{n,3}(s)\big) = C_3(s_3) = a\, i_{1,3}(\omega), \qquad a \neq 0.$$

Since $\mathrm{pr}_{n,2} = \mathrm{pr}_{3,2}\mathrm{pr}_{n,3}$ and $\mathrm{pr}_{3,2} i_{1,3} = 0$, it follows that $\mathrm{pr}_{n,2}(C_n(s)) = 0$. Applying i) with $m = 2$ and with its index n replaced by $n-2$, we obtain $C_n(s) = i_{n-2,n}(u)$ for some $u \in M_{n-2}$. Finally, the restriction of $\mathrm{pr}_{n,3}$ to the subbundle $\mathscr{E}_{n-2} \subset \mathscr{E}_n$ is the composite

$$\mathscr{E}_{n-2} \xrightarrow{\mathrm{pr}_{n-2,1}} \mathscr{E}_1 \xrightarrow{i_{1,3}} \mathscr{E}_3.$$

Consequently

$$\mathrm{pr}_{n,3}\, i_{n-2,n} = i_{1,3}\mathrm{pr}_{n-2,1}.$$

Comparison with the preceding display gives $\mathrm{pr}_{n-2,1}(u) = a\omega$. Taking $s' = a^{-1}u$ proves ii). □

Proposition 4.3.3. *Under* (C)*, the order of Cartier nilpotency of $\mathscr{E}_n$ is at least $\lfloor (n+1)/2 \rfloor$ for every $n \geq 1$.*

Proof. Choose $s_n \in M_n$ mapping to ω. Repeatedly applying Lemma 4.3.2 ii), with the index decreasing by two, gives

$$C_n^j(s_n) = a_j\, i_{n-2j,n}(s_{n-2j}), \qquad a_j \neq 0, \quad 0 \leq j \leq \left\lfloor \frac{n-1}{2} \right\rfloor,$$

where every s_{n-2j} maps to ω; set $i_{n,n} = \mathrm{id}$. The scalars remain non-zero under the p^{-1}-linear operator C_n. At the last step the lift lies in M_1 or M_2, and is non-zero. Hence $C_n^{\lfloor (n-1)/2 \rfloor}(s_n) \neq 0$, which proves the bound. □

Proposition 4.3.4. *Assume* (A) *and* (C)*. Then for the Frobenius invariant vector bundles $\mathscr{E}_r$ with inductive limit $\mathscr{E}$ constructed above, and with $E := \mathscr{E}|_x$, the comparison map*

$$\delta^1_{\mathscr{E}} : \mathrm{H}^1(\pi(X), E) = \varinjlim_r \mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E}_r) \longrightarrow \mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E})$$

is not bijective. In fact, the target space is infinite dimensional while the source space is zero.

Proof. As observed above, the semisimple parts $\mathrm{H}^1(X, \mathscr{E}_r)_{\mathrm{ss}}$ are one-dimensional and are killed by the transition maps. Since $\mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E}_r) = \mathrm{H}^1(X, \mathscr{E}_r)_{\mathrm{ss}}$, we obtain

$$\varinjlim_r \mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E}_r) = \varinjlim_r \mathrm{H}^1(X, \mathscr{E}_r)_{\mathrm{ss}} = 0.$$

We now analyze the inverse system occurring in stratified cohomology. Using the compatible Frobenius isomorphisms σ_r fixed in the construction, represent $\mathscr{E}_r$ by the F-divided system

$$\mathscr{E}_r^{(n)} = \mathscr{E}_r \qquad (n \geq 0),$$

whose transition maps are σ_r^{-1}. Passing to the filtered direct limit over r, we obtain

$$\mathscr{E}^{(n)} \simeq \mathscr{E} \qquad \text{for every } n,$$

compatibly with the Frobenius transition maps.

Write

$$\Phi_r := \mathrm{H}^1(X,\sigma_r^{-1}) \circ F_X^* : \mathrm{H}^1(X,\mathscr{E}_r) \longrightarrow \mathrm{H}^1(X,\mathscr{E}_r)$$

for the p-linear cohomological Frobenius determined by σ_r, and put

$$N_r := \mathrm{H}^1(X,\mathscr{E}_r)_{\mathrm{nil}}.$$

Thus the subscript nil refers to the nilpotent part for Φ_r. The kernel of $\mathrm{H}^1(X,\mathscr{E}_r) \to \mathrm{H}^1(X,\mathscr{E}_{r+1})$ is the semisimple line spanned by the extension class. Hence $N_r \subset N_{r+1}$; Riemann–Roch and $h^0(X,\mathscr{E}_r) = 1$ give $\dim N_r = r$. Set $N_0 = 0$. Since every transition map kills the semisimple line whereas its restriction to N_r is injective, the filtered direct limit identifies with

$$\mathrm{H}^1(X,\mathscr{E}) = \varinjlim_r \mathrm{H}^1(X,\mathscr{E}_r) = \bigcup_r N_r =: N,$$

and N is infinite dimensional. Compatibility of the σ_r induces a p-linear endomorphism $\Phi : N \to N$ whose restriction to N_r is Φ_r.

We next identify Φ on the graded pieces of N. Put $V_q = \mathbb{F}_p[t]/(t^q)$. On the dual $\mathbb{Z}_p$-module, the element t acts as

$$(1+t)^{-1} - 1 = -\frac{t}{1+t} = -t + t^2 - \cdots.$$

Since this polynomial has a non-zero linear term, its action is again one nilpotent Jordan block of length q. We may therefore choose a $\mathbb{Z}_p$-equivariant self-duality $\beta_q : V_q \xrightarrow{\sim} V_q^\vee$. Being a morphism of $\mathbb{F}_p$-local systems, it induces a Frobenius-compatible self-duality $\beta_q : \mathscr{E}_q \xrightarrow{\sim} \mathscr{E}_q^\vee$.

We use this self-duality only in the fixed rank q; no compatibility between the choices for different q is needed. It exchanges the inclusions and projections in the tower. More precisely, when $q = m + n$, the dual of

$$0 \longrightarrow \mathscr{E}_n \longrightarrow \mathscr{E}_q \longrightarrow \mathscr{E}_m \longrightarrow 0$$

is identified via β_q, and the induced identifications of the end terms, with the reversed sequence

$$0 \longrightarrow \mathscr{E}_m \longrightarrow \mathscr{E}_q \longrightarrow \mathscr{E}_n \longrightarrow 0.$$

Indeed, each Jordan block has a unique submodule of every dimension.

Use β_q in Serre duality. Since Φ_q and C_q are adjoint, the nilpotent parts are in perfect duality:

$$\langle\,,\,\rangle_q : N_q \times M_q \longrightarrow k.$$

We regard N_j and M_j as subspaces of N_q and M_q, respectively, using the maps in the tower. Naturality of Serre duality, together with the reversal of the two flags under the chosen self-duality, gives

$$(N_j)^\perp = M_{q-j} \qquad (0 \le j \le q).$$

Consequently

$$(N_j/N_{j-1})^\vee \simeq M_{q-j+1}/M_{q-j} \qquad (1 \le j \le q).$$

Now fix $r \ge 1$ and take $q = r+2$. For $j = q$ and $j = r$, respectively, this gives

$$(N_{r+2}/N_{r+1})^\vee \simeq M_1, \qquad (N_r/N_{r-1})^\vee \simeq M_3/M_2.$$

Since cohomological Frobenius and Cartier are adjoint, the transpose of the map induced by Φ is precisely $\overline{C}_3 : M_3/M_2 \xrightarrow{\sim} M_1$, which is an isomorphism by (C). Therefore

$$\Phi : N_{r+2}/N_{r+1} \xrightarrow{\sim} N_r/N_{r-1} \qquad (r \geq 1).$$

These graded isomorphisms imply $\Phi : N \twoheadrightarrow N$. To see this, argue by induction on r. Given $v \in N_r$, choose $w_0 \in N_{r+2}$ such that $\Phi(w_0) \equiv v \pmod{N_{r-1}}$ by the preceding isomorphism. The difference $v - \Phi(w_0)$ lies in N_{r-1} and therefore has a preimage by the induction hypothesis.

By the exact sequence recalled in Subsection 2.2.1, we have

$$0 \longrightarrow \varprojlim{}^1_n \mathrm{H}^0(X, \mathscr{E}^{(n)}) \longrightarrow \mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E}) \longrightarrow \varprojlim_n \mathrm{H}^1(X, \mathscr{E}^{(n)}) \longrightarrow 0.$$

For every n, we have $\mathrm{H}^0(X, \mathscr{E}^{(n)}) \simeq k$, and the transition maps are bijective p-linear maps on k. Therefore $\varprojlim^1_n \mathrm{H}^0(X, \mathscr{E}^{(n)}) = 0$, and we obtain a natural isomorphism

$$\mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E}) \xrightarrow{\sim} \varprojlim_n \mathrm{H}^1(X, \mathscr{E}^{(n)}).$$

Under the compatible identifications above, the inverse system on the right is

$$N \xleftarrow{\Phi} N \xleftarrow{\Phi} N \xleftarrow{\Phi} \cdots.$$

Since $\Phi : N \twoheadrightarrow N$, the projection

$$\varprojlim_{\Phi} N \longrightarrow N, \qquad (v_0, v_1, \ldots) \longmapsto v_0,$$

is surjective. Hence $\varprojlim_\Phi N$, and therefore $\mathrm{H}^1_{\mathrm{str}}(X, \mathscr{E})$, is infinite dimensional. □

4.4. **Explicit computation for hyperelliptic curves of genus 2 over a field of characteristic 3.** We now verify (C) on explicit curves in characteristic three. The characteristic is used only in this realization; the arguments of Section 4.3 require (A) and (C). Let X be a hyperelliptic curve of genus 2 with two marked Weierstrass points, and $\mathrm{char}(k) = 3$. By fixing two points, such curve can be defined by the following affine covering: $X = U_0 \cup U_1$, where

$$U_0 = X \setminus \{\infty\} : y^2 = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x;$$
$$U_1 = X \setminus \{0\} : w^2 = a_1v^5 + a_2v^4 + a_3v^3 + a_4v^2 + a_5v,$$

and on the intersection $U_{01} := U_0 \cap U_1$, we have the following relations: $y = wv^{-3}$ and $x = v^{-1}$. We first provide the necessary and sufficient conditions for X to have Hasse-Witt rank 1.

Lemma 4.4.1. *Let X be the curve obtained by patching the two affine curves*

$$U_0 = X \setminus \{\infty\} : y^2 = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x;$$
$$U_1 = X \setminus \{0\} : w^2 = a_1v^5 + a_2v^4 + a_3v^3 + a_4v^2 + a_5v,$$

subject to the relations: $y = wv^{-3}$ and $x = v^{-1}$. Then X is smooth (hence has genus 2 and is hyperelliptic) and has Hasse-Witt rank one if and only if

$$\begin{cases} a_4a_2 = a_5a_1 \\ a_1a_5(a_2^4 + a_1^3a_5) \neq 0 \\ a_3(a_2a_5^2 + a_4^3 - a_3a_4a_5) \neq 0. \end{cases}$$

The proof will be given in the Appendix A.1. □

Lemma 4.4.2 (The vector bundle $\mathscr{E}_2$)**.** *Assume the conditions of Lemma 4.4.1. The cohomology class of the 1-cocycle $e_{01} \in \mathscr{O}(U_{01})$*

$$e_{01} = (a+bx)x^{-2}y = a.\frac{w}{v} + b.\frac{y}{x}, \tag{8}$$

is invariant under Frobenius if $(a,b) \neq (0,0)$ satisfies:

$$\begin{bmatrix} a_4 & a_1 \\ a_5 & a_2 \end{bmatrix} . \begin{bmatrix} a^3 \\ b^3 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}. \tag{9}$$

The proof will be given in the Appendix A.2. □

Lemma 4.4.3. *For the bundle $\mathscr{E}_2$ of Lemma 4.4.2, the induced Cartier operator vanishes on M_2; its nilpotency order is one.*

The proof will be given in the Appendix A.3. □

Example 4.4.4. *If $a_i = -1$ for all $1 \leq i \leq 5$, then we can choose*

$$\omega = \frac{(x-1)dx}{y}; \quad \eta = \frac{(x+1)dx}{y}$$
$$e_{01} = (1+x)\frac{y}{x^2} = \frac{w}{v} + \frac{w}{v^2}.$$

Lemma 4.4.5 (The vector bundle $\mathscr{E}_3$)**.** *The bundle $\mathscr{E}_3$ is given by a $GL_3(\mathscr{O}(U_{01}))$-cocycle of the form:*

$$A_3 = \begin{bmatrix} 1 & e_{01} & f_{01} \\ 0 & 1 & e_{01} \\ 0 & 0 & 1 \end{bmatrix}.$$

with $f_{01} = -e_{01}^2$.

The proof will be given in the Appendix A.4. □

After the change of trivialization in Appendix A.4, we use the cocycle (12), transporting the Frobenius isomorphism with it. Then each element in $\mathrm{H}^0(X, \Omega_X \otimes \mathscr{E}_3)$ is a pair of vectors of local sections subject to the relation

$$\begin{bmatrix} {}^3s_0 \\ {}^2s_0 \\ {}^1s_0 \end{bmatrix} = \begin{bmatrix} 1 & e_{01} & 0 \\ 0 & 1 & e_{01} \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} {}^3s_1 \\ {}^2s_1 \\ {}^1s_1 \end{bmatrix}$$

where ${}^js_i \in \Omega(U_i)$ for $i = 0,1$ and $j = 1,2,3$, that is

$$\begin{cases} {}^3s_0 - {}^3s_1 = e_{01}.{}^2s_1 \\ {}^2s_0 - {}^2s_1 = e_{01}.{}^1s_1 \\ {}^1s_0 - {}^1s_1 = 0. \end{cases}$$

Since ${}^1s_0 - {}^1s_1 = 0$, these local sections (of the sheaf Ω_X) glue together to obtain a global section s in $\Omega(X)$.

Our finding here is the following.

Lemma 4.4.6. *Assume the conditions of Lemma 4.4.1 and use the bundles $\mathscr{E}_2, \mathscr{E}_3$ constructed above. Then condition* (C) *holds. In particular, the Cartier nilpotency order of $\mathscr{E}_3$ is two.*

The proof is given in Appendix A.5. □

Together with Propositions 4.3.3 and 4.3.4, this verifies the counterexample in characteristic three.

Example 4.4.7. (i) *The curve $y^2 = -x^5 - x^4 - x^3 - x^2 - x$, defined over $\mathbb{F}_3$ and base-changed to k, satisfies Lemma 4.4.1, hence* (C).

(ii) *The smooth curve $y^2 = -x^5 + x^4 + x^3 - x^2 + x$ has p-rank zero: here $a_2^4 + a_1^3 a_5 = 0$, and its non-zero Cartier operator satisfies $C^2 = 0$. Thus it does not satisfy* (A) *and cannot be used in this construction.*

APPENDIX A.

A.1. **Proof of Lemma 4.4.1.** X is smooth if and only if the degree-five polynomial $f(x)$ has no repeated roots. This is the same as saying that $a_1 a_5 \neq 0$ and the discriminant of the quartic polynomial $a_5 x^4 + a_4 x^3 + a_3 x^2 + a_2 x + a_1$,

$$\begin{aligned}\Delta := {} & a_5^3 a_1^3 + a_5^2 a_3^2 a_1^2 + a_5 a_4 a_3^2 a_2 a_1 + a_5 a_3^4 a_1 \\ & - a_5 a_3^3 a_2^2 - a_4^3 a_2^3 - a_4^2 a_3^3 a_1 + a_4^2 a_3^2 a_2^2 \neq 0.\end{aligned}$$

The space of global differential 1-forms $\Omega(X)$ is spanned by the holomorphic differential

$$\omega_1 = y^{-1} dx \quad \text{and} \quad \omega_2 = y^{-1} x dx.$$

The Cartier operator in this basis is given by:

$$C = \begin{bmatrix} a_2^{1/3} & a_1^{1/3} \\ a_5^{1/3} & a_4^{1/3} \end{bmatrix}.$$

For X to have Hasse-Witt rank one, it is necessary that $\det(C) = 0$, that is

$$a_1 a_5 = a_2 a_4.$$

Under this equality, the condition $\Delta \neq 0$ can be expressed as

$$a_3(a_2 a_5^2 + a_4^3 - a_3 a_4 a_5) \neq 0.$$

Since C has rank one, its stable rank is one exactly when $C^2 \neq 0$. In that case there is a basis $\{\omega, \eta\}$ of $\Omega(X)$ with $C(\omega) = 0$ and $C(\eta) = \eta$. We can choose

$$\omega = a_4 \omega_1 - a_5 \omega_2.$$

If we present η in the form $\eta = \lambda_1^3 \omega_1 + \lambda_2^3 \omega_2$ then

$$C(\eta) = C(\lambda_1^3 \omega_1 + \lambda_2^3 \omega_2) = \lambda_1 C(\omega_1) + \lambda_2 C(\omega_2).$$

So λ_1, λ_2 need to satisfy:

$$\begin{bmatrix} a_2^{1/3} & a_1^{1/3} \\ a_5^{1/3} & a_4^{1/3} \end{bmatrix} \cdot \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} \lambda_1^3 \\ \lambda_2^3 \end{bmatrix}$$

Elementary computation shows that the existence of $\eta \neq 0$ is equivalent to

$$a_2^4 + a_1^3 a_5 \neq 0.$$

A.2. **Proof of Lemma 4.4.2.** One needs to check $e_{01}^3 - e_{01} = 0$ in $\mathrm{H}^1(X, \mathscr{O}_X)$. Indeed, we have

$$\begin{aligned}
e_{01}^3 &= a^3\frac{w^3}{v^3} + b^3\frac{y^3}{x^3} \\
&= a^3\frac{w(a_1v^5 + a_2v^4 + a_3v^3 + a_4v^2 + a_5v^1)}{v^3} + b^3\frac{y(a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x^1)}{x^3} \\
&= a^3w(a_1v^2 + a_2v + a_3) + b^3y(a_5x^2 + a_4x + a_3) + \big(a^3a_4\frac{w}{v} + a^3a_5\frac{w}{v^2} + b^3a_2\frac{y}{x} + b^3a_1\frac{y}{x^2}\big) \\
&= f_1 + f_0 + \big(a^3a_4\frac{w}{v} + a^3a_5\frac{y}{x} + b^3a_2\frac{y}{x} + b^3a_1\frac{w}{v}\big) \\
&= f_1 + f_0 + \big(\frac{w}{v}(a^3a_4 + b^3a_1) + \frac{y}{x}(a^3a_5 + b^3a_2)\big) \\
&= f_1 + f_0 + \big(\frac{w}{v}.a + \frac{y}{x}.b\big) \quad \text{by (9)} \\
&= f_1 + f_0 + e_{01}
\end{aligned}$$

where $f_0 = b^3y(a_5x^2 + a_4x + a_3) \in \mathscr{O}(U_0)$, and $f_1 = a^3w(a_1v^2 + a_2v + a_3) \in \mathscr{O}(U_1)$.

A.3. **Proof of Lemma 4.4.3.** We now compute the image of $\begin{bmatrix} {}^2s_i \\ \omega \end{bmatrix} \in \mathrm{H}^0(X, \Omega_X \otimes F^*\mathscr{E}_2)$ under C, where recall that $\omega = a_4\omega_1 - a_5\omega_2$. According to (8), $e_{01}^3 = \frac{y^3}{x^6}(a^3 + b^3x^3)$. Hence

$$\begin{aligned}
e_{01}^3.\omega = \frac{y^3}{x^6}(a^3 + b^3x^3).\omega &= \frac{y^2}{x^6}(a^3 + b^3x^3).(a_4 - a_5x)\,dx \\
&= \frac{(b^3x^3 + a^3)(a_4 - a_5x)(a_5x^4 + a_4x^3 + a_3x^2 + a_2x + a_1)}{x^5}dx \\
&= \big(-b^3a_5^2x^3 + b^3(a_4^2 - a_5a_3)x + b^3(a_4a_3 - a_5a_2) - a^3a_5^2\big)dx \\
&\quad - \big(b^3a_4a_1 + a^3(a_4^2 - a_5a_3) + a^3(a_4a_3 - a_5a_2)v + a^3a_4a_1v^3\big)dv.
\end{aligned}$$

Consequently

$$\begin{aligned}
{}^2s_0 - {}^2s_1 = e_{01}^3.\omega &= \big(-b^3a_5^2x^3 + b^3(a_4^2 - a_5a_3)x + b^3(a_4a_3 - a_5a_2) - a^3a_5^2\big)dx \\
&\quad - \big(b^3a_4a_1 + a^3(a_4^2 - a_5a_3) + a^3(a_4a_3 - a_5a_2)v + a^3a_4a_1v^3\big)dv.
\end{aligned}$$

So, up to an element of $\Omega(X)$, we have that

$$^2s_0 = \big(-b^3a_5^2x^3 + b^3(a_4^2 - a_5a_3)x + b^3(a_4a_3 - a_5a_2) - a^3a_5^2\big)dx \tag{10}$$

$$^2s_1 = \big(b^3a_4a_1 + a^3(a_4^2 - a_5a_3) + a^3(a_4a_3 - a_5a_2)v + a^3a_4a_1v^3\big)dv. \tag{11}$$

With these choices, $C({}^2s_0) = C({}^2s_1) = 0$. This lift of ω, together with the section $[\omega, 0]^t$, gives two independent elements in the kernel of the raw Cartier map. The section $[\eta, 0]^t$ is not in its kernel. Since $h^0(X, \Omega_X \otimes \mathscr{E}_2) = 3$ and $\dim M_2 = 2$, the kernel is exactly the nilpotent part (transported by the chosen Frobenius isomorphism). Hence $C_2(M_2) = 0$.

A.4. **Proof of Lemma 4.4.5.** We have

$$e_{01}^3 - e_{01} = f_0 + f_1$$

with $f_0 = b^3 y(a_5x^2 + a_4x + a_3) \in \mathcal{O}(U_0)$ and $f_1 = a^3 w(a_1v^2 + a_2v + a_3) \in \mathcal{O}(U_1)$. Hence

$$\begin{bmatrix} 1 & f_0 & -f_0^2 \\ 0 & 1 & f_0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & e_{01} & -e_{01}^2 \\ 0 & 1 & e_{01} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & f_1 & -f_1^2 \\ 0 & 1 & f_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & e_{01}^3 & -(f_1+f_0)^2 - 2e_{01}(f_0+f_1) - e_{01}^2 \\ 0 & 1 & e_{01}^3 \\ 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & e_{01}^3 & -e_{01}^6 \\ 0 & 1 & e_{01}^3 \\ 0 & 0 & 1 \end{bmatrix}.$$

To pass to a simpler cocycle, write

$$e_{01}^2 = \frac{(a+bx)^2 f(x)}{x^4} = g_0 - g_1, \qquad g_i \in \mathcal{O}(U_i),$$

by splitting the Laurent polynomial into its non-negative and negative powers of x. If E_{13} denotes the elementary matrix, then

$$(I + g_0E_{13})A_3(I + g_1E_{13})^{-1} = \begin{bmatrix} 1 & e_{01} & 0 \\ 0 & 1 & e_{01} \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus we may use the equivalent cocycle

$$\begin{bmatrix} 1 & e_{01} & 0 \\ 0 & 1 & e_{01} \\ 0 & 0 & 1 \end{bmatrix}, \tag{12}$$

with its transported Frobenius isomorphism. The changes of trivialization preserve the flag and the quotient coordinates.

A.5. **Proof of Lemma 4.4.6.** We use the cocycle (12). Thus a section of $\Omega_X \otimes F^*\mathscr{E}_3$ with last coordinate ω satisfies

$${}^2s_0 - {}^2s_1 = e_{01}^3\omega, \qquad {}^3s_0 - {}^3s_1 = e_{01}^3\,{}^2s_1.$$

Choose ${}^2s_i$ as in (10)–(11), and put

$$c_0 = b^3a_4a_1 + a^3(a_4^2 - a_5a_3),$$
$$c_1 = a^3(a_4a_3 - a_5a_2), \qquad c_3 = a^3a_4a_1.$$

Then

$$e_{01}^3\,{}^2s_1 = \frac{(a_1v^4 + a_2v^3 + a_3v^2 + a_4v + a_5)^2}{wv^4}(a^3v^3 + b^3)(c_0 + c_1v + c_3v^3)\,dv.$$

Separating the negative powers of v and using $v^{-j}\,dv/w = -x^{j+1}\,dx/y$ gives

$$e_{01}^3\,{}^2s_1 = \tau_1 - P(x)\frac{dx}{y}, \qquad \tau_1 \in \Omega(U_1), \qquad P(x) = A_2x^2 + A_3x^3 + A_4x^4 + A_5x^5,$$

where

$$A_2 = a_5^2(a^3c_0 + b^3c_3) + b^3c_1(a_4^2 - a_3a_5) - b^3c_0(a_2a_5 + a_3a_4),$$
$$A_3 = b^3\big(-c_1a_4a_5 + c_0(a_4^2 - a_3a_5)\big),$$
$$A_4 = b^3(c_1a_5^2 - c_0a_4a_5), \qquad A_5 = b^3c_0a_5^2.$$

Consequently the correct local choices are

$${}^3s_0 = -P(x)\frac{dx}{y}, \qquad {}^3s_1 = -\tau_1.$$

They may both be changed by the same global form $\gamma \in \Omega(X)$. Since $C({}^2 s_i) = C(\omega) = 0$, their Cartier images glue to a global form. The only powers in $f(x)P(x)$ that contribute to Cartier are x^5 and x^8; their coefficients are

$$R_5 = a_3 A_2 + a_2 A_3 + a_1 A_4, \qquad R_8 = a_5 A_3 + a_4 A_4 + a_3 A_5.$$

Equation (9) and $a_1 a_5 = a_2 a_4$ give $(a,b) = t(a_4, a_5)$ for some $t \neq 0$. Substitution in these expressions yields

$$R_8 = 0, \qquad R_5 = t^6 a_3 a_4^2 a_5^2 K, \quad K = (a_5^2 a_2 + a_4^3)(a_5^2 a_2 + a_4^3 - a_3 a_4 a_5).$$

Both factors of K are non-zero by Lemma 4.4.1: for the first, use $a_2^3(a_5^2 a_2 + a_4^3) = a_5^2(a_2^4 + a_1^3 a_5)$. Thus $R_5 \neq 0$, and the raw Cartier map gives

$$C({}^3 s_0) = C({}^3 s_1) = -R_5^{1/3}\omega_2 + C(\gamma), \qquad \omega_2 = \frac{x\,dx}{y}. \tag{13}$$

Here $C(\gamma) \in k\eta$, whereas $\omega_2 \notin k\eta$: the image line of the Cartier matrix in Appendix A.1 has both coordinates non-zero. Hence (13) has a non-zero nilpotent component.

Finally take the nilpotent component of the constructed section in $\mathrm{H}^0(X, \Omega_X \otimes F^*\mathscr{E}_3)$, using the transported Frobenius isomorphism. Its projection to $F^*\mathscr{E}_2$ is unchanged, since that projection lies in the Cartier kernel by Appendix A.3. Its image under the raw Cartier map is therefore a non-zero multiple of $[\omega, 0, 0]^t$. Transporting back gives a lift $s_3 \in M_3$ of ω with $C_3(s_3) = a\, i_{1,3}(\omega)$, $a \neq 0$. Together with $C_2(M_2) = 0$ and Lemma 4.3.2 i), this proves (C).

A.6. **An example with vanishing first stratified cohomology.** We give an example, on a hyperelliptic curve in characteristic 3, of an F-periodic line bundle whose first stratified cohomology group vanishes.

Let X be any smooth curve given by the equations of Section 4.4; we do not impose Hasse–Witt rank one in this subsection. Let $\mathscr{L} := \mathscr{O}(\infty - O)$ be the line bundle corresponding to the Weierstrass divisor $\infty - O$, where O is $(x,y) = (0,0)$ on U_0 and ∞ is $(v,w) = (0,0)$ on U_1. Since $2\cdot\infty - 2\cdot O$ is the divisor of x^{-1}, $\mathscr{L}^{\otimes 2}$ is trivial. Hence $\mathscr{L}^3 \cong \mathscr{L}$, that is $\mathscr{L}$ is an F-periodic line bundle. Thus, $\mathscr{L}$ determines a stratified line bundle which will also be denoted by $\mathscr{L}$.

To compute $\mathrm{H}^1_{\mathrm{str}}(X, \mathscr{L})$, we study Frobenius on $\mathrm{H}^1(X, \mathscr{L})$, or equivalently Cartier on

$$\mathrm{H}^0(X, \Omega_X \otimes \mathscr{L}^\vee) = \mathrm{H}^0(X, \Omega_X(O - \infty)).$$

$\mathscr{L}$ is non-trivial: otherwise a rational function with divisor $\infty - O$ would give a degree-one map $X \to \mathbb{P}^1$, contrary to $g(X) = 2$. Riemann–Roch therefore gives $h^0(X, \Omega_X(O-\infty)) = 1$, so there is a unique (up to scalars) differential form which has a pole at O of order at most 1, vanishes at ∞ and is regular elsewhere. Using the notations of Section 4.4, the form is:

$$\omega_1 = \frac{dx}{y},$$

and the Cartier map on $\mathrm{H}^0(X, \Omega(O-\infty))$ is the following composition (cf. (6)):

$$C_\sigma : \mathrm{H}^0(X, \Omega_X(O-\infty)) \xrightarrow{\cdot x^{-1}} \mathrm{H}^0(X, \Omega_X(3\cdot(O-\infty))) \xrightarrow{C} \mathrm{H}^0(X, \Omega_X(O-\infty)). \tag{14}$$

The image of ω_1 under this map is

$$C_\sigma(\omega_1) = C\left(\frac{y^2x^2dx}{x^3y^3}\right) = \frac{1}{xy}\cdot C((a_5x^5+a_4x^4+a_3x^3+a_2x^2+a_1x)x^2dx)$$
$$= \frac{1}{y}\cdot a_3^{1/3}dx = a_3^{1/3}\cdot\omega_1.$$

We conclude that $\mathrm{H}^1_{\mathrm{str}}(X,\mathscr{L}) = 0$ if and only if the coefficient a_3 is 0. Notice that, according to the proof of 4.4.1, if $a_3 = 0$ then X has to be ordinary. Indeed, if $a_3 = 0$, $\Delta = (a_5a_1 - a_4a_2)^3$. Thus, if X is smooth, its Cartier matrix is invertible and X is ordinary. For example, $y^2 = x^5 + x$ over $\overline{\mathbb{F}}_3$ is smooth and has $a_3 = 0$, giving the claimed example.

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